

Anesthesia AIIMS 2017

Question 1:

A 40 year old female with burns over her abdomen, both limbs and back and presented after 8 hours. Which of the following is the formula for calculation of fluid infusion?

- a) 2 ml/kg X %TBSA for the next 16 hours
- b) 5 ml/kg X %TSBA for the next 16 hours
- c) 4 ml/kg X %TSBA for first 8 hours then 2 ml/kg X %TBSA for next 16 hours
- d) 4 ml/kg X%TSBA for the next 16 hours

Question 2:

A 35-year-old woman was brought to the casualty ward with a history of road traffic accident (RTA) and head trauma. Her eye opens to pain, moans, can localize limb towards the pain stimulus on the left hand and away from the pain stimulus on the right hand and both the legs in extended posture. Calculate her GCS.

- a) 11
- b) 9
- c) 7
- d) 3

Question 3:

A patient who was on ventilator and being ventilated for past few days suddenly pulls out of the endotracheal tube. What is the next step of management?

- a) Re intubate
- b) Oxygenate and reintubate
- c) Start CPR
- d) Watch for spontaneous breaths and reintubate if needed

Question 4:

The time required for preoxygenation before tracheal intubation?

- a) 1 min
- b) 3 min
- c) 5-7 min
- d) 5 min

Question 5:

Nerve fibre least susceptible to local anesthetic is:

- a) A-delta
- b) A- gamma
- c) A- beta
- d) C

Question 6:

Which of the following can be used as an intravenous induction agent?

- a) Lorazepam
- b) Bupivacaine
- c) Dexmedetomidine
- d) Neostigmine

Question 7:

Which of the following is the correct sequence?

- a) Assess the patient's response, Activate ERS, check carotid pulse, Start CPR
- b) Check carotid pulse, Start CPR, Call for help and activate ERS, Defibrillate
- c) Defibrillate, Assess response, Check carotid pulse, Maintain airway
- d) Start CPR, Activate ERS, Defibrillate, Check pulse

Answer Key

Question No.	Correct Option
1	a
2	b
3	d
4	b
5	d
6	a
7	a

Detailed Explanations

Solution to Question 1:

Fluid resuscitation in burns:

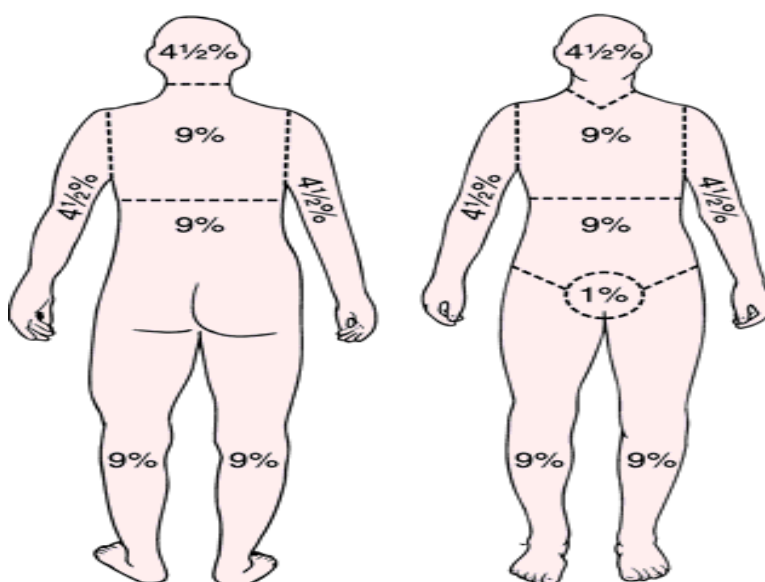
According to ATLS 10th edition, 2018, the resuscitation formula used in thermal burns is $2 \text{ mL/kg} \times \% \text{ TBSA}$.

First 8 hours: $1 \text{ mL/kg} \times \% \text{ TBSA}$ (lactated Ringer solution)

Next 16 hours: $1 \text{ mL/kg} \times \% \text{ TBSA}$ (lactated Ringer solution)

For a patient presenting 8 hours after sustaining thermal burns, the total volume using resuscitation formula should be given in the next 16 hours i.e; $2 \text{ mL/kg} \times \% \text{ TBSA}$ in the next 16 hours while maintaining target urine output of 0.5 mL/kg/hr for adults.

In adults, urine output should be maintained between 30 and 50 cc/hr to minimize potential over-resuscitation.



Note:

The goal of resuscitation is to maintain the fine balance of adequate perfusion as indicated by urine output. It is important to understand that formulas provide a starting target rate; subsequently, the amount of fluids provided should be adjusted based on a urine output target of 0.5 mL/kg/hr for adults and 1 mL/kg/hr for children weighing less than 30 kg.

Total percentage of the burnt area is calculated clinically using the rule of nines.

Solution to Question 2:

The GCS (Glasgow Coma Scale) of this patient is E2V2M5, which comes to a total score of 9.

E2-Eye opens to pain

V2-Moans

M5-Localizes to pain on the left hand (the score represents the best performance elicited, which would be localizing towards the pain stimulus)

Revised Glasgow coma scale for head injury:

Minimum GCS is 3, maximum GCS is 15.

Solution to Question 3:

About half of self-extubated patients may not require re-intubation. Pre-extubation FiO₂, PaO₂/FiO₂, and ventilator mode, as well as gender, may be useful in determining the likelihood that a self-extubated patient will require re-intubation and also watch for spontaneous breathing.

Solution to Question 4:

Preoxygenation with tidal volume breathing of 3 mins is required before tracheal intubation.

Preoxygenation or denitrogenation is the process of replacing nitrogen in the lungs with oxygen.

This delays the onset of critical hypoxia during the period of apnea by 5-8 minutes.

This lengthened apnea time provides an improved margin of safety while the anesthesiologist secures the airway and resumes ventilation.

Solution to Question 5:

The nerve fibre least susceptible to local anaesthetic is type C. The order of susceptibility of nerve fibres to local anaesthetics is B > A-δ > A-γ > A-β > A-α > C i.e

- B fibres are the most susceptible/least resistant
- C fibres are the least susceptible/most resistant

Factors that increase the sensitivity of nerve fibres are :

- Smaller diameter
- Unmyelinated nerves
- Anatomical and physiological factors

*Note: Although theoretically, C fibres are supposed to be the most susceptible to local anaesthetics, practically it was found to be the least susceptible.

Solution to Question 6:

Lorazepam is a benzodiazepine that can be used for induction of anesthesia at a dose of 0.1mg/kg, although midazolam is preferred among benzodiazepines.

Benzodiazepines are mainly used for sedation and less commonly for induction of anesthesia due to the availability of better induction agents (propofol, etomidate, ketamine, and thiopentone).

Dexmedetomidine is a selective alpha-2 agonist and is used primarily as an adjuvant. So, a better option would be lorazepam.

Solution to Question 7:

BLS sequence:

First step – Assess safety of the surroundings.

Second step - Assess for responsiveness

Third step – Proceed if the patient is unresponsive. Shout for help.

If the patient is unresponsive assess for breathing and carotid pulse simultaneously for no more than 10 seconds.

Anesthesia AIIMS 2018

Question 1:

A patient is undergoing modified radical neck dissection (MRND) for laryngeal malignancy. While ligating the internal jugular vein, his etCO₂ dropped from 40 mmHg to 14 mmHg, and the patient developed hypotension along with cardiac arrhythmia. Which of the following is the most likely cause?

- a) Sympathetic overactivity
- b) Vagal stimulation
- c) Venous air embolism
- d) Carotid body stimulation

Question 2:

At what level does the spinal cord end in adults?

- a) L3-L4
- b) L1-L2
- c) T12-L1
- d) L2-L3

Question 3:

Which of the following is not a contraindication to the use of morphine?

- a) Bronchial asthma
- b) Renal failure
- c) Emphysema
- d) Acute myocardial infarction

Question 4:

Cardiac output can be determined using the following except?

- a) Thermal dilution technique

- b) Capnography
- c) Esophageal doppler
- d) Echocardiography

Answer Key

Question No.	Correct Option
1	c
2	b
3	d
4	b

Detailed Explanations

Solution to Question 1:

A drastic fall in etCO₂ with hypotension during neck surgery is most likely due to venous air embolism (VAE).

In modified radical neck dissection (MRND), a head-up position is given to provide a bloodless surgical field. VAE into severed vessels is likely to occur when the surgical site is above the right atrium and venous pressure at the incision site is below the atmospheric pressure. Pulmonary artery pressure will rise and etCO₂ will fall after a venous air embolism.

In sympathetic overactivity and carotid body stimulation, blood pressure will rise.

In vagal stimulation, a drastic drop in etCO₂ is unlikely.

Solution to Question 2:

The distal termination of the spinal cord is the lower border of L1 in adults.

Because the spinal cord ends at L1, lower nerve roots course some distance before exiting the intervertebral foramina. These lower spinal nerves form the cauda equina.

Note:

In children, the spinal cord ends at L3 and moves up with age.

Solution to Question 3:

Acute myocardial infarction is not a contraindication for morphine use.

Morphine is the opioid of choice for myocardial infarction as it relieves anxiety, chest pain, and myocardial oxygen demand.

Contraindications to the use of Morphine:

- Bronchial asthma: Histamine release can cause bronchoconstriction and precipitate attacks of asthma.
- Renal failure: Morphine-6-glucuronide (metabolite) can accumulate and lead to prolonged respiratory depression.
- Emphysema: Respiratory depression and decreased response to hypoxia can cause hypercapnia in patients with COPD
- Head injury: Hypercapnia due to respiratory depression can increase ICP. Miosis, vomiting, and effect on wakefulness make interpreting the signs of head injury difficult.
- Biliary colic: Constriction of sphincter of Oddi can worsen biliary spasm
- Hypotension and shock

Solution to Question 4:

Capnography is a measurement of end-tidal CO₂, which is used to confirm tracheal intubation and assess ventilation in the perioperative period.

Cardiac output measurements technique:

- Thermodilution
- Dye dilution, using indocyanine green
- Lithium dilution
- Pulse contour devices
- Esophageal doppler
- Thoracic bioimpedance
- Echocardiography

Note: Thermal dilution method uses a non-toxic liquid of temperature different from the blood. This fluid is injected into a vein and is detected on the arterial side using a catheter with a temperature sensor. The rate of flow of this liquid is used to calculate the cardiac output.

Anesthesia AIIMS 2019 (May & Nov)

Question 1:

Which is not an obvious advantage of high pressure oxygen through nasal cannula:

- a) Decreases need of intubation
- b) Reduces nasopharyngeal dead space
- c) Heating and humidification
- d) PEEP

Question 2:

Duration of action of Proparacaine:

- a) 2 min
- b) 5 min
- c) 40 min
- d) 20 min

Question 3:

Glycopyrrolate is used as a pre-anesthetic agent for:

- a) Reducing the secretions
- b) Sedative effect
- c) Skeletal muscle relaxant
- d) Anxiolytic

Question 4:

Identify the device shown in the image below.



- a) Automated External Desynchronizer
- b) Automated External Defibrillator
- c) Automated Electronic Defibrillator
- d) Automated External decompressor

Question 5:

You are an intern posted in the ICU. While taking a blood sample for ABG, which of the following steps would you not do?

- a) Normal ph can't rule out acid base disorder
- b) Add 0.5 ml Heparin to the syringe before taking blood
- c) If modified Allens test is negative, alternate site is chosen
- d) Radial artery is the preferred site

Question 6:

Which of the following is not a risk factor for postoperative pulmonary complications?

- a) BMI $\geq$ 30
- b) Well controlled asthma
- c) Upper abdominal surgery
- d) Age $\geq$ 70 years

Question 7:

Which of the following is associated with respiratory alkalosis?

- a) Assisted control mode ventilation
- b) Controlled mandatory
- c) Pressure controlled
- d) SIMV

Answer Key

Question No.	Correct Option
1	a
2	d
3	a
4	b
5	b
6	b
7	a

Detailed Explanations

Solution to Question 1:

High flow nasal cannula (HFNC), which delivers oxygen at high flow, and pressure through the nasal cannula.

Advantages of high-flow nasal cannula:

- Soft and pliable nasal prongs provide comfort to the patient.
- Warming and humidification of secretions: The heating of air to core temperature and humidification of the inspired air is better at high-flow. The humidification causes increased water content in the secretions, which facilitates the removal of secretions. It also decreases the work of breathing and avoids airway dessication and epithelial injury.
- Reduces dead space: The high flow of the oxygen delivered clears the upper airway of the expired air, which reduces dead space by a decrease of rebreathing and allows a higher fraction of minute ventilation to participate in alveolar gas exchange.

- Continuous positive airway pressure (CPAP): HFNC increases the positive end-expiratory pressure (PEEP), and reduces the work of breathing.
- High flow rate causes decreased entrainment of room air, allowing more accurate delivery of oxygen.

Solution to Question 2:

The duration of action of proparacaine is approximately 10-20 minutes.

Proparacaine hydrochloride ophthalmic solution is used as a topical anesthetic agent.

Indications: For procedures which require corneal anesthesia of short duration such as:

- Tonometry
- Gonioscopy
- Removal of corneal foreign bodies
- For short corneal and conjunctival procedures

Contraindication: Patients with known hypersensitivity

Solution to Question 3:

Glycopyrrolate is used as a pre-anesthetic agent for reducing the secretions.

Anticholinergic drugs e.g. atropine, glycopyrrolate are routinely administered with anticholinesterases in order to attenuate the undesirable muscarinic effects. Glycopyrrolate is usually preferred because it has less vagolytic effects than atropine at doses that inhibit secretions and does not cross the blood-brain barrier. It should be administered as early as possible to maximize its effectiveness.

Glycopyrrolate 0.2 mg IV, is administered to dry airway secretions before flexible scope intubation of the trachea, during the bronchoscopic examination.

Solution to Question 4:

The device is an Automated External Defibrillator (AED).

This is a special medical device that is commonly used for recognizing the absence/presence of rapid ventricular tachycardia or ventricular fibrillation. With no need for intervention by an operator, it determines whether an electrical shock right to the heart must be performed (defibrillation).

An automated external defibrillator can be used on those victims, who are unconscious and aren't breathing in a normal way. As a rule, it is used together with CPR in cases of sudden cardiac arrest. It is forbidden to use it on patients who are less than eight years old or those weighing less than 25 kgs.

Solution to Question 5:

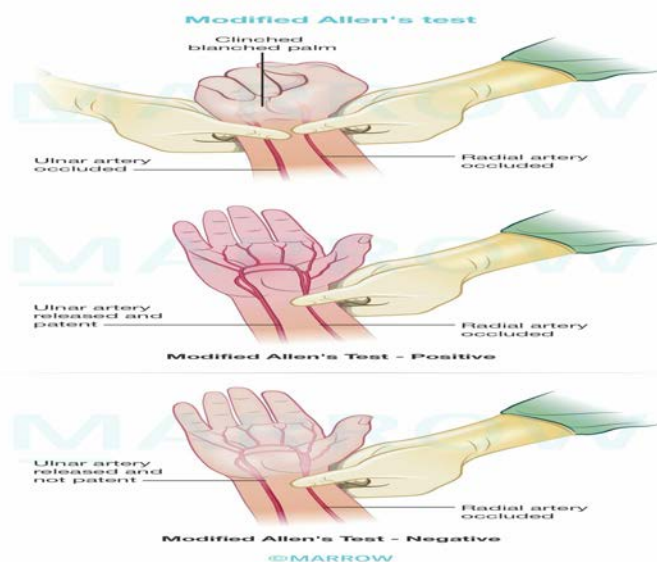
0.5 ml of Heparin is not added to the syringe before taking blood.

Very little heparin is needed in the sample to prevent clotting; 0.05 to 0.10 ml of a dilute solution will help prevent coagulation of 1 ml of blood without affecting its pH. This is achieved by drawing 0.5 ml into the syringe and flushing it out. The residue that remains adherent to the syringe is sufficient for the procedure.

Heparin must be added to the syringe as an anticoagulant. Because the pH of heparin is near 7.0, and the P_{O_2} and P_{CO_2} of the heparin solution are near room air values, excess heparin can alter ABG measurements. After flushing the syringe with heparin, a sufficient amount usually remains in the dead space of the syringe and needle for anticoagulation without distortion of the ABG determination.

Note:

- Modified Allen's test positive indicates that the collateral blood flow is normal
- Modified Allen's test negative indicates that the collateral blood flow is abnormal and an alternate site must be chosen.



Solution to Question 6:

Well-controlled asthma is not a risk factor for postoperative pulmonary complications.

Risk factors for postoperative pulmonary complications:

- Age >70 years
- COPD (uncontrolled)
- Neck, upper abdominal, Aortic, neurologic surgical procedures

- Prolonged procedure >2 hours
- General anesthesia with intubation
- BMI > 30
- Albumin concentration < 35 g/L
- Inability to walk two blocks on level land or one flight of stairs

Solution to Question 7:

Respiratory alkalosis can occur in Assist Control Mode Ventilation

Ventilatory Mode	ACMV(Assist Control Mode Ventilation)	IMV(Intermittent Mandatory Ventilation)	PSV(Pressure Support Ventilation)	NIV(Noninvasive Ventilation)
Variables set by User (Independent)	Tidal volumeVentilator rateFio ₂ PEEP levelPressure limit	Tidal volumeMandatory ventilator rateFio ₂ PEEP levelPressure limitSpontaneous breaths between assisted breaths	Inspiratory pressure levelFio ₂ PEEPPressure limit	Inspiratory and expiratory pressure levelFio ₂
Variables Monitored by the user (Dependent)	Peak, mean, and plateau airway pressuresVEABGI/E ratio	Peak, mean, and plateau airway pressuresVEABGI/E ratio	Tidal volumeRespiratory rateVEABG	Tidal volumeRespiratory rateVEABG
Trigger cycle limit	Patient effortTimer Pressure limit	Patient effortTimer Pressure limit	Pressure limitInspiratory flow	Pressure limitInspiratory flow
Advantages	Patient controlGuaranteed ventilation	Patient controlComfort from spontaneous breathsGuaranteed ventilation	Patient controlComfortAssures synchrony	Patient control
Disadvantages	Potential hyperventilation(Leading to respiratory alkalosis) Barotrauma and volutraumaEvery effective breath generates a ventilator volume	Potential dysynchronyPotential hypoventilation	No timer backupPotential hypoventilation	Mask interface may cause discomfort and facial bruisingLeaks are commonHypoventilation

Anesthesia AIIMS 2020 (May and Nov INI-CET)

Question 1:

A patient of blunt chest trauma has been started on mechanical ventilation. Which of the following is not correct about the initial setting?

- a) Respiratory rate - 12 Breaths/ min
- b) PEEP - 5cm H₂O
- c) FiO₂- 100%
- d) Tidal volume - 12ml/kg of actual BW

Question 2:

Glycopyrrolate is used as a pre-anesthetic agent for:

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- b) Sedative effect
- c) Skeletal muscle relaxant
- d) Anxiolytic

Question 3:

Cardiac output can be determined using the following except?

- a) Thermal dilution technique
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- c) Esophageal doppler
- d) Echocardiography

Question 4:

Identify the equipment shown in the image below.



- a) Mechanical ventilator
- b) Infusion pump
- c) CPAP machine
- d) High flow nasal cannula

Question 5:

The blue pipeline in the operation theatre is used for delivering which of the following gases?

- a) Oxygen
- b) Nitrous oxide
- c) Air
- d) Vacuum

Question 6:

The given set of instruments are used in:



- a) Central line insertion
- b) Arterial line insertion
- c) Airway management
- d) Umbilical catheter insertion

Question 7:

On your morning walk in the park, you notice an unconscious man lying on the ground. Which of the following options would be the next best step?

- a) Check carotid pulse, start CPR, and call for help; Defibrillate if no pulse.
- b) Call for help, check pulse, do CPR, defibrillate if no pulse
- c) Observe
- d) Ignore and continue to walk

Question 8:

Which of the following can be considered as effective CPR?

- a) 1, 2, 3, and 4
- b) 1, 2, and 4
- c) 2, 3, and 4
- d) 1 and 2 only

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	d
5	b
6	c
7	b
8	c

Detailed Explanations

Solution to Question 1:

The tidal volume of 12ml/kg of actual body weight is an incorrect initial setting for mechanical ventilation.

Initial ventilatory settings are as given below:

- Tidal volume - 4-6 ml/kg of IBW (Ideal Body Weight)
- Respiratory rate - 12 Breaths/ min
- FiO₂ - 100% to begin with and titrated to lesser values to prevent oxygen toxicity
- PEEP - Begin with 5 cm H₂O, titrate to 10 cm H₂O (Carefully increased as high PEEP may impair ventricular filling in a patient with blunt chest trauma)

Solution to Question 2:

Glycopyrrolate is used as a pre-anesthetic agent for reducing the secretions.

Anticholinergic drugs e.g. atropine, glycopyrrolate are routinely administered with anticholinesterases in order to attenuate the undesirable muscarinic effects. Glycopyrrolate is usually preferred because it has less vagolytic effects than atropine at doses that inhibit secretions and does not cross the blood-brain barrier. It should be administered as early as possible to maximize its effectiveness.

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Solution to Question 3:

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Cardiac output measurements technique:

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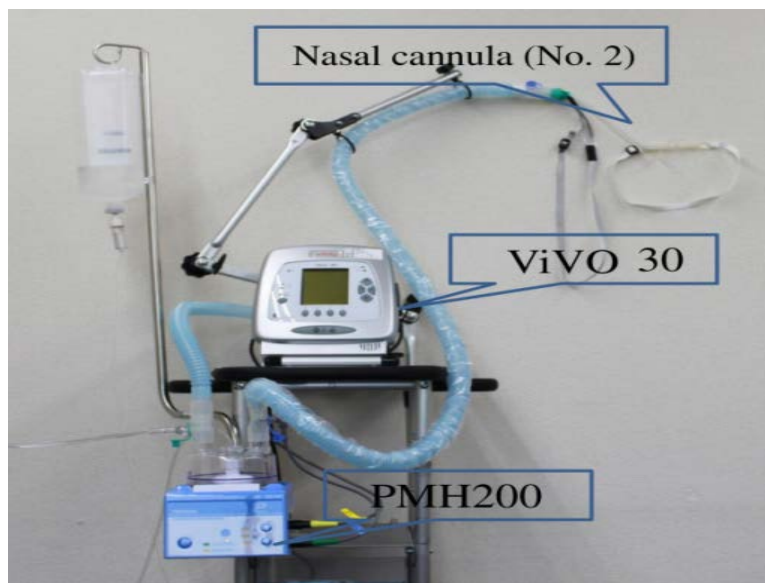
Solution to Question 4:

The equipment shown in the image is high flow nasal cannula.

High flow nasal cannula is a mode of respiratory support used in the management of acute respiratory failure. It delivers humidified gas via the nasal cannula at high flow rates.

- HFNC can deliver oxygen at 40L/min, 37°C, and 99.9% relative humidity.
- The high-flow oxygen is delivered directly to the nasopharynx. This produces FiO₂ equal to that delivered by traditional mask devices.
- The benefits of HFNC include reduced work of breathing, improved gas exchange by nasopharyngeal dead-space washout, and decreased energy required to humidify and heat respiratory gases.

The HFNC equipment shown in the image below uses a heated humidifier (named PMH200), a nasal cannula (no. 2), and a ventilator (brand: vivo30).

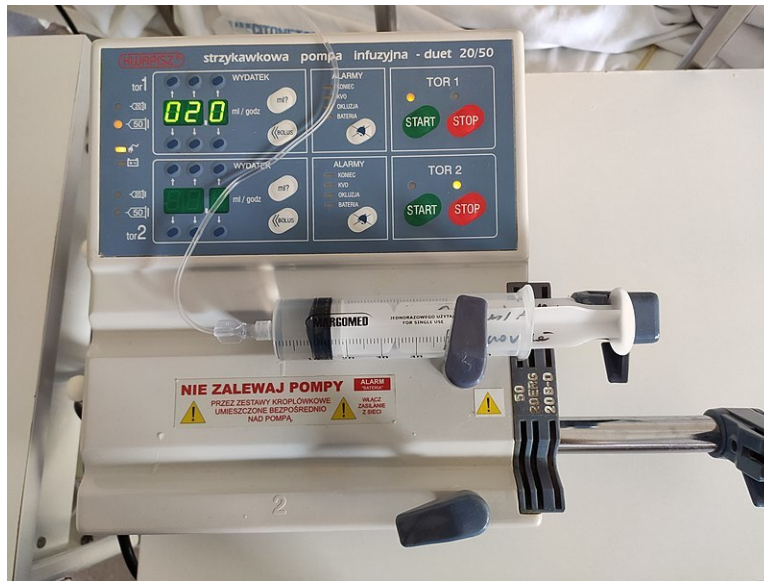


Other options:

Option a: Mechanical ventilator



Option b: Infusion Pump



Option c: CPAP



Solution to Question 5:

The blue pipeline in operation theatre is for nitrous oxide.

Other options:

Oxygen: White

Air: Black

Vacuum/Central suction: Yellow

Solution to Question 6:

The image shows a tray in an airway cart with the instruments necessary for airway management.

An airway cart may include:

- Laryngoscope blades of various sizes, including a rigid fiberoptic.
- Tracheal tubes of various sizes, and tracheal tube guides such as semirigid stylets, ventilating tube changer, light wands, and forceps that are required to manipulate the distal portion of the tracheal tube.
- Masks of various sizes.
- Laryngeal mask airways of various sizes.
- Flexible fiberoptic intubation equipment.
- Retrograde intubation equipment.
- At least one device to perform emergency noninvasive airway ventilation such as esophageal tracheal tube, a hollow jet ventilation stylet.
- Equipment required for emergency invasive airway access.
- An exhaled CO₂ detector.

Solution to Question 7:

The next best step is to Call for help, check pulse, do CPR, defibrillate If no pulse.

Steps in basic life support:

First step – Assess safety of the surroundings.

Second step - Assess for responsiveness

Third step – Proceed if the patient is unresponsive. Shout for help.

If the patient is unresponsive, assess for breathing and carotid pulse simultaneously for no more than 10 seconds.

In the given scenario, it is already mentioned that you are in a park (scene safety has been assessed), and that the person is unconscious (responsiveness has been assessed). So, the immediate next step is to call for help, and start CPR.

Solution to Question 8:

The ideal rescue breath rate for adults is 10 breaths/min, while in infants and children up to puberty it is 20-30 breaths/min. As the person's age group is not specified in the question, assuming an adult, statement 1 is incorrect, while 2, 3, and 4 are correct.

Effective chest compression in adults:

- Provide 100-120 compressions per minute (30 compressions for every 15-18 sec).

- Place your arms, one over the other in the midline, on the lower 1/3rd of the patient's sternum, and lock your arms.
- Use both arms to press the chest to a depth of 5-6 cm (2 - 2.4 inches) or more.
- Press hard and fast
- Allow for full chest recoil after every compression.

Effective chest compressions in a child older than 1 year and up to puberty:

- The compression rate is 100 - 120/min (30 compressions for every 15-18 sec).
- One or both palms are placed in the midline.
- The depth of compression is 2 inches.
- Press hard and fast
- Allow for full chest recoil after every compression.

Effective chest compressions in infants (<1 year):

- 100 to 120 compressions per minute (30 compressions for every 15-18 sec).
- One provider: Place two fingers, one between the nipple line and the other 1 cm below.
- Two providers: Encircle the infant's torso with both hands with both thumbs pointing cephalic positioned 1cm below the nipples over the sternum.
- Depth of compressions should be at least 1.5 inches or 1/3 the depth of the infant's chest.
- Press hard and fast
- Allow for full chest recoil after every compression.

Anesthesia NEET 2018

Question 1:

What is the mechanism of action of the curare group of muscle relaxants?

- a) Persistently depolarizing at neuromuscular junction
- b) Competitively blocking the binding of ACh to its receptors
- c) Repetitive stimulation of ACh receptors on muscle endplate
- d) Inhibiting the calcium channels on presynaptic membrane

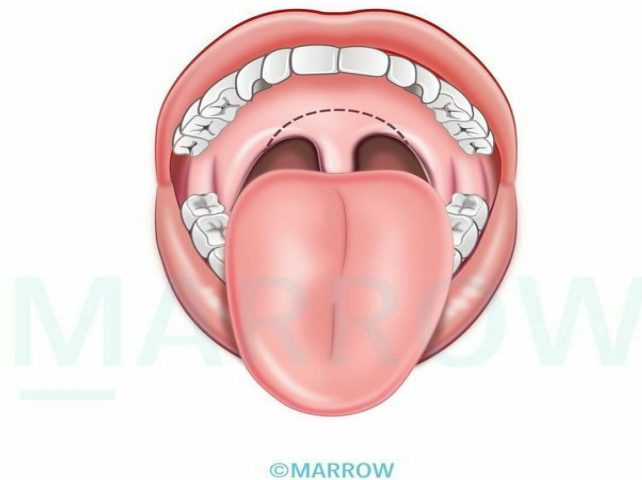
Question 2:

Which of the following nerves is popularly used in clinical anaesthesia to monitor neuromuscular blockade?

- a) Ulnar Nerve
- b) Median Nerve
- c) Radial nerve
- d) Mandibular nerve

Question 3:

Identify the Mallampatti class of this patient.



- a) Class II
- b) Class I
- c) Class IV
- d) Class III

Question 4:

In which of the following devices is Murphy's eye seen?

- a) Nasopharyngeal airway
- b) Laryngoscope
- c) Bronchoscope
- d) Endotracheal tube

Question 5:

What is the most effective circuit for anaesthesia under spontaneous breathing?

- a) Mapleson A
- b) Mapleson B
- c) Mapleson C
- d) Mapleson D

Question 6:

The best type of anesthesia for pre eclampsia is:

- a) Spinal anesthesia
- b) Epidural anesthesia
- c) General anesthesia
- d) Caudal block

Question 7:

What alternative drug would you use for cardiac arrest if epinephrine is not effective?

- a) Vasopressin
- b) Atropine
- c) Amiodarone
- d) Adenosine

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	d
5	a
6	b
7	c

Detailed Explanations

Solution to Question 1:

The curare group of drugs includes non-depolarizing neuromuscular blocking agents that act by competitively blocking the binding of acetylcholine to its receptors.

This leads to a decrease in the effect of acetylcholine and thus, neuronal transmission to the muscle does not occur.

Solution to Question 2:

Ulnar nerve stimulation at wrist is popularly used to assess the adequacy of neuromuscular blockade.

Neuromuscular blockade is measured using a peripheral nerve stimulator.

Nerve	Response
Ulnar nerve	Contraction of adductor pollicis

Nerve	Response
Facial nerve	Contraction of orbicularis oculi
Posterior tibial nerve	Contraction of flexor hallucis brevis

Solution to Question 3:

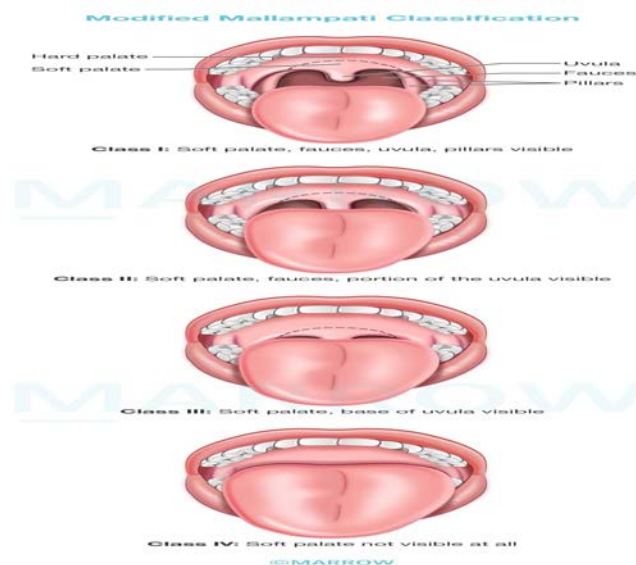
The image shows Mallampatti class II airway.

The modified Mallampatti classification is used in the assessment of airway. It is a predictor of the difficulty of intubation or difficult airway.

The assessment is performed with the patient sitting up straight, mouth open and tongue maximally protruded, without saying "ahh."

Modified Mallampatti classification:

- Class 0: Ability to see any part of the epiglottis.
- Class I: Soft palate, fauces, uvula visible.
- Class II: Soft palate and fauces seen, tip of uvula not visible.
- Class III: Soft palate, base of uvula visible.
- Class IV: Soft palate not visible at all. Only hard palate visible.



Classes 3 and 4 are difficult intubation situations.

Initially, the Mallampatti scoring system (1985) had only 3 classes. This was modified by Samsoon and Young to include class 4. Later class 0 was also added.

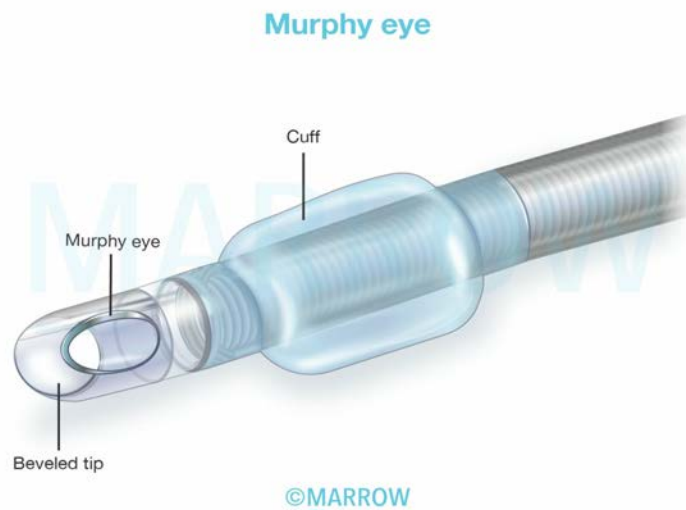


Solution to Question 4:

Murphy's eye is seen in the endotracheal tube (ETT).

Murphy's eye refers to the distal opening in the side wall of the ETT.

If the distal end of the lumen of the tube is blocked by secretions, Murphy's eye serves to provide an alternate route for ventilation.



Solution to Question 5:

The most effective circuit for anesthesia under spontaneous breathing is Mapleson A or Magill's circuit.

When the Mapleson A circuit is used for spontaneous ventilation, the fresh gas flow rate of the circuit is closest to the physiological minute ventilation of the patient. Hence it is preferred over other circuits.

Solution to Question 6:

Continuous epidural anesthesia is the first choice for most patients with preeclampsia during labor, vaginal delivery, and cesarean section.

Pre-eclampsia patients have a risk of severe airway edema, which makes intubation difficult. Thus continuous epidural anesthesia is advantageous as it avoids the need for intubation. Besides, continuous epidural anesthesia can improve uteroplacental perfusion and also decrease catecholamine secretion.

But one major prerequisite for regional anesthesia in preeclamptic patients is normal platelet count and normal coagulation. Coagulopathy is a contraindication for regional anesthesia.

Minimal platelet count for regional anesthesia is 100,000/ μ L, but some cases can be taken up with a platelet count as low as 70,000/ μ L.

General anesthesia is indicated in patients with coagulopathy and in severe pre-eclampsia with complications.

Spinal anesthesia can be given in mild pre-eclampsia but hypotension associated with spinal anesthesia limits its use in moderate to severe pre-eclampsia.

Since the severity of pre-eclampsia or the type of delivery (vaginal or cesarean) is not mentioned, epidural anesthesia has an advantage over other techniques for above-mentioned reasons. Hence the best option here would be epidural anesthesia.

Solution to Question 7:

Epinephrine and amiodarone are the two main drugs approved by the AHA for the resuscitation of a cardiac arrest patient.

The first dose of epinephrine 1 mg IV is administered after the 2nd defibrillator shock is delivered and it is continued every 3-5 minutes.

If this is not effective, 300 mg IV bolus dose of amiodarone can be given and another dose of 150 mg given if necessary

Anesthesia NEET 2019

Question 1:

What is the mechanism of action of the curare group of muscle relaxants?

- a) Persistently depolarizing at neuromuscular junction
- b) Competitively blocking the binding of ACh to its receptors
- c) Repetitive stimulation of ACh receptors on muscle endplate
- d) Inhibiting the calcium channels on presynaptic membrane

Question 2:

Which among the following IV anaesthetics does not cause myocardial depression?

- a) Etomidate
- b) Propofol
- c) Thiopentone
- d) Ketamine

Question 3:

Which of the following is not used for induction of anaesthesia in pediatric patients?

- a) Halothane
- b) Sevoflurane
- c) Desflurane
- d) Nitrous oxide

Question 4:

Which of the following agents does not elicit pain on injection?

- a) Ketamine
- b) Propofol
- c) Etomidate

d) Thiopentone

Question 5:

Which among the following is most commonly used to monitor the depth of anaesthesia?

- a) EEG
- b) Provoked lower oesophageal contractility
- c) Entropy
- d) Bispectral index

Question 6:

Which of the following is measured by the device shown in the given image?



- a) Oxygen saturation
- b) Oxygen content of the blood
- c) Partial pressure of oxygen
- d) Amount of inspired oxygen

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	a
5	d
6	a

Detailed Explanations

Solution to Question 1:

The curare group of drugs includes non-depolarizing neuromuscular blocking agents that act by competitively blocking the binding of acetylcholine to its receptors.

This leads to a decrease in the effect of acetylcholine and thus, neuronal transmission to the muscle does not occur.

Solution to Question 2:

Etomidate, a short-acting I.V. anesthetic agent doesn't cause myocardial depression.

Etomidate is the most cardiac stable intravenous agent due to its lack of effect on the sympathetic nervous system and on the function of the baroreceptor. Hence, etomidate is the I.V. agent of choice in patients with coronary artery disease, cardiomyopathy, and cerebral vascular disease.

Side effects of Etomidate:

- The highest incidence of nausea and vomiting among inducing agents
- Superficial thrombophlebitis and pain on injection
- Adrenocortical suppression

Note: Although Ketamine has an indirect sympathetic stimulatory effect, it possesses a direct negative inotropic effect on the myocardium. Hence, ketamine has to be used cautiously in critically ill patients with shock, as they are prone to have a depletion of endogenous catecholamines. Ketamine infusion in such patients can unmask its direct negative inotropic effects and further decrease the blood pressure and cardiac output.

Hence, for this MCQ, Etomidate is the best answer.

Solution to Question 3:

Desflurane is a highly pungent gas and can produce laryngospasm if used for inhalational induction. Hence it is not used as an induction agent.

The incidence of laryngospasm following desflurane induction is almost 50%. It is usually used for maintenance of anaesthesia only and is of particular value in obese patients, neurosurgical and spinal fusion procedures where rapid awakening is required for neurological function assessment.

Option A: Halothane is hepatotoxic but can be used (Sevoflurane is preferred)

Option B: Sevoflurane is the preferred inhalational agent for induction in the pediatric age group. Cardiovascular depression, bradycardia, and arrhythmias are less frequent with sevoflurane than with halothane and hence the former is preferred.

Option D: Nitrous oxide has the highest minimum alveolar concentration and so it is the least potent. It is used in combination with other potent inhaled anaesthetics to reduce their dose. It can be used to hasten the induction of anaesthesia along with other induction agents.

Solution to Question 4:

Ketamine does not elicit pain on injection, instead produces profound analgesia.

Propofol is formulated as a lipid emulsion, which causes pain on injection. A new aqueous prodrug, fospropofol is associated with less pain on injection.

Etomidate, although it causes pain on injection, the lipid formulation of etomidate is associated with a much less frequent incidence of pain.

Thiopentone at a concentration greater than 2.5% can produce pain on injection and venous thrombosis.

Solution to Question 5:

The bispectral index is the most commonly used method to monitor the depth of anaesthesia.

Note: Middle latency auditory evoked potential is the most sensitive indicator to evaluate the depth of anaesthesia.

Solution to Question 6:

The given image depicts a pulse oximeter. Pulse oximetry is used for monitoring pulse rate and oxygen saturation of a patient.

A small sensor is placed on the finger, toe, earlobe, or another convenient place. The relative absorption of light by oxyhemoglobin (HbO₂) and deoxyhemoglobin is assessed by measuring the amount of red and infrared light emerging from tissues traversed by light rays and processed by the device, producing an oxygen saturation level.

Pulse oximetry does not measure the absolute content or the partial pressures of oxygen or carbon dioxide.

A pulse oximeter is not always reliable as it reads:

- Falsely high - carboxyhemoglobinemia
- Falsely low - severe anemia, low cardiac output states, hypovolemia, increased systemic vascular resistance
- 80% to 85%, irrespective of the arterial oxygen saturation - methemoglobinemia

Anesthesia NEET 2020

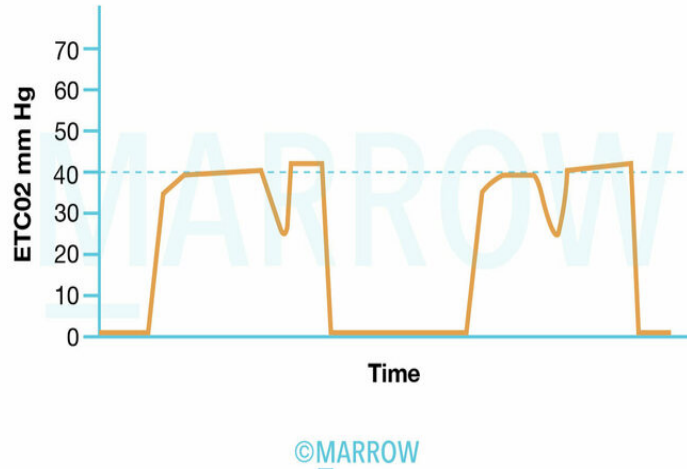
Question 1:

Which of the following investigations provides the most accurate prognostic information with respect to predicting risks of perioperative cardiac complications?

- a) Exercise ECG testing
- b) Dobutamine stress echocardiography
- c) Myocardial perfusion scintigraphy
- d) Coronary angiography

Question 2:

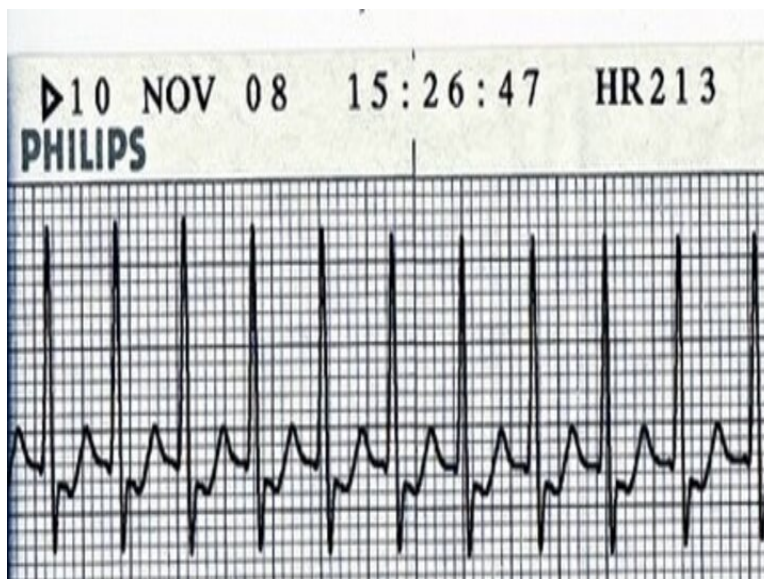
A capnograph obtained from an intubated patient undergoing controlled ventilation is given below. Which of the following is a likely cause of the finding seen?



- a) Esophageal intubation
- b) Inspiration with cardiac oscillations
- c) Bronchospasm
- d) Spontaneous respiratory effort

Question 3:

A 50-year-old patient with a history of hypertension and diabetes mellitus developed the following rhythm. His blood pressure was 90/40 mm Hg and he had a feeble pulse. What is the next step of management?



- a) IV adenosine
- b) IV diltiazem
- c) DC cardioversion
- d) Ibutilide

Question 4:

A patient is given a nicotinic receptor antagonist as a muscle relaxant. Which drug is given postoperatively to recover from muscle weakness?

- a) Physostigmine
- b) Neostigmine
- c) Carbachol
- d) Succinylcholine

Answer Key

Question No.	Correct Option
1	b

2	d
3	c
4	b

Detailed Explanations

Solution to Question 1:

Among the given investigations, dobutamine stress echocardiography provides the most accurate prognostic information with respect to predicting risks of perioperative cardiac complications.

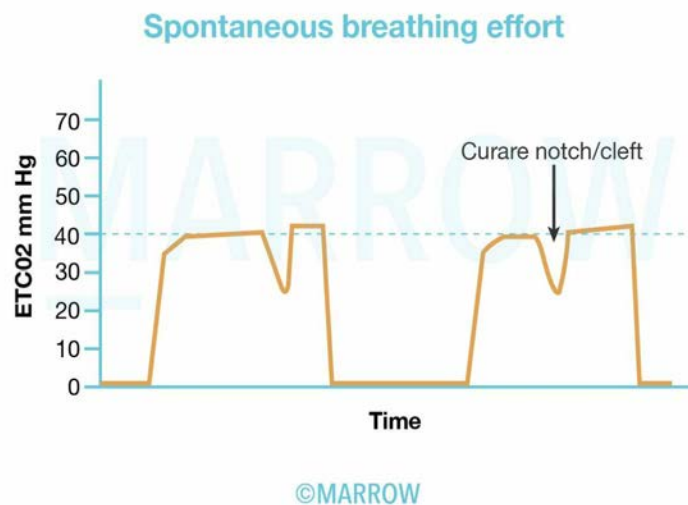
Preoperative cardiac stress testing can help identify the presence and severity of preexisting CAD. The prognostic performance of these tests is better assessed based on positive and negative likelihood ratios. Among all the given options, dobutamine stress echocardiography has the maximum positive likelihood ratio and hence provides the most accurate prognostic information.

Even though coronary angiography is considered the gold standard, it may not absolutely risk-stratify patients because many perioperative cardiac events may be related to noncritical coronary lesions.

Solution to Question 2:

The curare notch in the plateau phase of the capnograph indicates that the patient is trying to breathe spontaneously. The patient's inspiratory effort causes some fresh gas to be sucked in the capnometer, resulting in a lower carbon dioxide concentration and a notch.

Image below shows a capnograph with clefts during phase III which indicates spontaneous breathing efforts by the patient during controlled mechanical ventilation. It is also known as a curare notch or cleft.



Solution to Question 3:

The clinical scenario suggests supraventricular tachycardia with hemodynamic instability and the next step of management is DC cardioversion (50-100J).

If the patient was hemodynamically stable, initial treatment would be vagal maneuvers (Valsalva maneuver) in awake patients. This is because vagal maneuvers alone can terminate approximately 20% to 25% of reentry supraventricular tachycardias (SVTs).

If this is unsuccessful, then iv adenosine 6 mg is given. If necessary another dose of adenosine 12 mg may be administered after 1-2 minutes.

If the drug is administered via a central venous catheter, then these doses are reduced to 3 mg and 6 mg respectively.

Solution to Question 4:

Neostigmine is given postoperatively to recover from muscle weakness.

Non-depolarizing muscle relaxants (NDMRs) act mainly by the competitive antagonism of nicotinic (Nm) receptors. Neostigmine inhibits acetylcholinesterase and inhibits the degradation of acetylcholine. The increased accumulation of undegraded acetylcholine can effectively compete with NDMRs, and thereby displace them from the site and antagonize their effects.

Anesthesia INI-CET 2021

Question 1:

Ventilatory support is required if SpO₂ becomes less than 90% because

- a) Tissues become vulnerable for hypoxia below 90%.
- b) Peripheral chemoreceptors becomes insensitive to hypoxia below 90%.
- c) Central chemoreceptors becomes insensitive to hypoxia below 90%.
- d) Patient capacity to generate adequate tidal volume decreases below 90%.

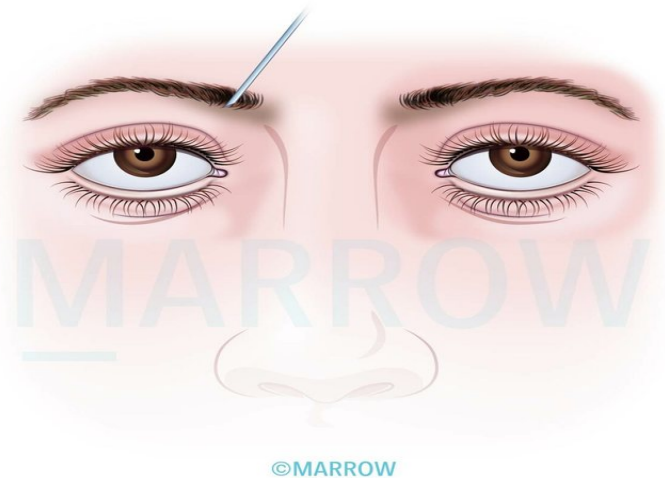
Question 2:

In conventional oxygen therapy, which device will deliver highest FiO₂?

- a) Venturi mask
- b) Hudson mask
- c) Non rebreathing mask
- d) Nasal cannula

Question 3:

What is the nerve blocked in the following local anesthesia technique?



- a) Nasociliary nerve
- b) Anterior ethmoidal nerve
- c) Sphenopalatine nerve
- d) Greater palatine nerve

Question 4:

How long should elective surgery be deferred for a patient who had cough and fever, after he has recovered from COVID-19, and had not required any hospitalization?

- a) 6 weeks
- b) 8-10 weeks
- c) 12 weeks
- d) 4 weeks

Question 5:

A 42-year-old man who was under home isolation after testing positive for COVID-19, develops breathing difficulty and is brought to the ER. The patient is alert and responsive, but is breathing rapidly. SpO₂ is measured to be 86%. He is administered oxygen via a face mask with a reservoir bag with a flow rate of 10 L/min. After a trial of one hour, his SpO₂ is 88%. What is the next step in management?

- a) Increase flow rate to 15 L/min
- b) High-flow nasal cannula oxygen therapy
- c) Endotracheal intubation
- d) No specific intervention necessary

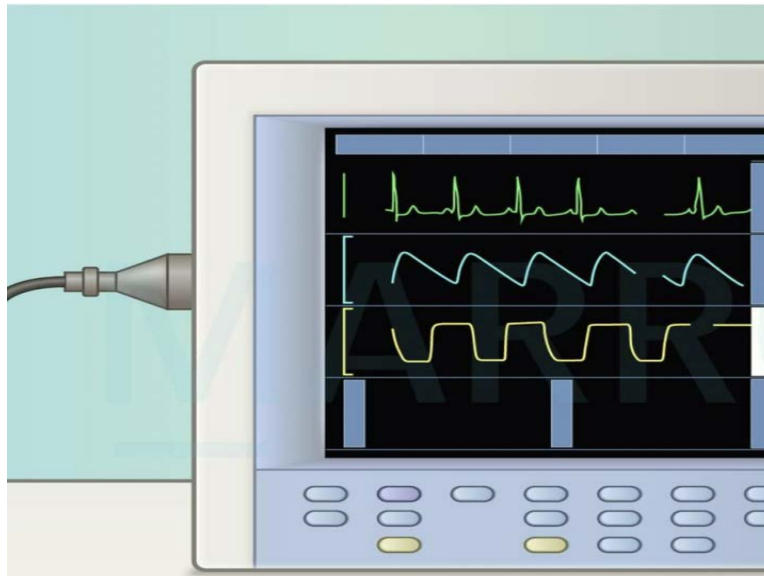
Question 6:

What is the mechanism of action of local anesthetics?

- a) Block sodium channels
- b) Block potassium channels
- c) Block calcium channels
- d) Block chloride channels

Question 7:

You are in the operating room and notice the tracing in yellow color on this device. What does it indicate?



- a) Capnography
- b) Airway pressure
- c) O₂ pressure in inhaled air
- d) O₂ pressure in exhaled air

Question 8:

What is the most common cause for hospital admission after day care surgery?

- a) Respiratory distress
- b) Post surgical pain
- c) Postoperative nausea and vomiting
- d) Bleeding

Question 9:

What is the frequency of tetanic stimulation used in neuromuscular monitoring?

- a) 0.2 Hz
- b) 2 Hz

- c) 20 Hz
- d) 50 Hz

Question 10:

A 60-year-old male patient was undergoing laparoscopic cholecystectomy. During the surgery, a sudden drop in EtCO₂ to 8 mmHg was noted. His SpO₂ became 90%, his blood pressure dropped to 80/50 mmHg, and peak airway pressure was 18 cm of H₂O. What is the diagnosis?

- a) Both 1 and 2
- b) Both 3 and 4
- c) 1 only
- d) 2 only

Question 11:

The maximum oxygen flow rate which can be kept with a conventional oxygen mask is

- a) 5 L/min
- b) 15 L/min
- c) 10 L/min
- d) 20 L/min

Question 12:

Which of the following can be used for induction of anesthesia, maintenance and can be safely used in endotracheal intubation?

- a) Halothane
- b) Desflurane
- c) Sevoflurane
- d) Isoflurane

Question 13:

During resuscitation, a patient received a 20% Intralipid injection. What could have been the most probable diagnosis?

- a) Local anaesthesia toxicity
- b) Asphyxia due to drowning
- c) Electrocution
- d) Pulseless electrical activity

Question 14:

A 50-year-old male patient has grade 3 ASA. Which of the following postbronchodilator PFT findings requires further evaluation of the lung function of the patient?

- a) $FEV_1 \leq 0.85$ and $FEV_1/FVC \leq 0.8$
- b) $FEV_1 \leq 0.85$ and $FEV_1/FVC \leq 0.75$
- c) $FEV_1 \leq 0.8$ and $FEV_1/FVC \leq 0.7$
- d) $FEV_1 \leq 0.8$ and $FEV_1/FVC \leq 0.75$

Answer Key

Question No.	Correct Option
1	a
2	c
3	a
4	a
5	b
6	a
7	a
8	c
9	d
10	c
11	c
12	c
13	a
14	c

Detailed Explanations

Solution to Question 1:

Tissues become vulnerable to hypoxia when SpO_2 falls below 90% and thus supplementary oxygen therapy is indicated below this oxygen saturation level.

In adults, children, and infants (older than 1 month), When SpO_2 falls below 90%, the PaO_2 becomes less than 60 mmHg and hence to prevent tissues from hypoxia, supplemental oxygen therapy should be started.

In neonates, when SpO_2 falls below 88%, the PaO_2 becomes less than 50 mmHg and hence supplemental oxygen therapy should be initiated in these patients.

Solution to Question 2:

Non rebreathing masks provide highest FiO_2 during conventional oxygen therapy.

The non-rebreather mask is a low flow device that can be used to deliver oxygen at higher concentrations (FiO_2 80-95%). It uses a reservoir bag to let the oxygen get concentrated while a one-way valve doesn't allow to breathe in expired air.

This type of masks are used in patients with significant hypoxia, with relatively normal ventilation, such as victims of trauma or carbon monoxide exposure.



Other options:

A) Venturi mask:

- Venturi mask is a high-flow, fixed-performance oxygen delivery system and the highest oxygen concentration (FiO₂) delivered is 60% at a flow rate of 15 L/min.
- It is based on the Bernoulli Principle, which states that in order to maintain a constant flow, a fluid must increase its velocity as it flows through a constriction.
- It is available in different colors, with the green valve venturi mask providing 60% FiO₂.
- This is useful with patients who require greater FiO₂ such as COPD patients, where it is important to provide controlled oxygenation.

Venturi masks



©Marrow

B) Hudson mask:

- Also known as simple face mask
- Low-flow, variable performance device
- They provide an oxygen flow rate of up to 5-10 L/min and provide a variable FiO₂.
- The body of the mask functions as reservoir for both oxygen and expired carbon dioxide.
- Used in patients who require oxygen therapy for shorter periods of time.

Hudson mask



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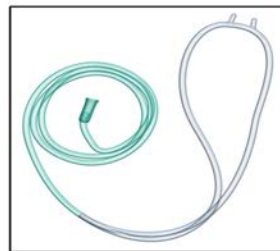
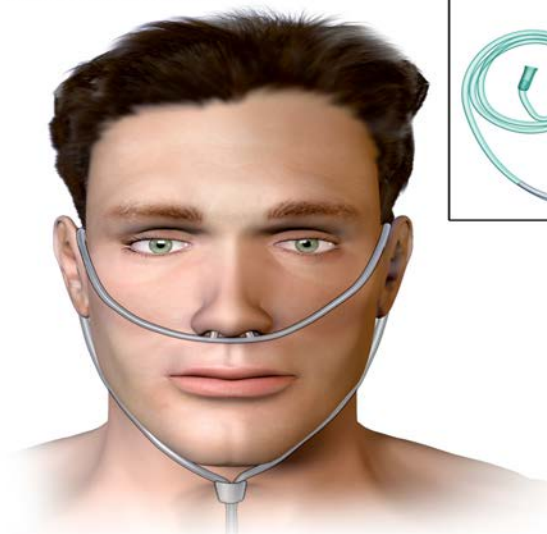
D) Nasal cannula:

- They are low-flow, variable performance devices, with flow rate of 1-4 L/min.
- Used in patients who require low supplemental oxygen.
- Nasopharynx acts as the reservoir in such cannulas.

Indications of a nasal cannula and other low-flow oxygen delivery systems include

- Minute ventilation $\leq 8-10$ L/min
- Breathing frequencies ≤ 20 breaths/min
- Normal inspiratory flow (10–30 L/min).

Nasal Cannula



Solution to Question 3:

The given image shows a nasociliary nerve block.

The procedure is done by inserting the needle 1 cm superior and 1.5 cm posterolateral to the medial canthal ligament of the eye.

The nasociliary nerve is a branch of the ophthalmic nerve which in turn is a branch of the trigeminal nerve.

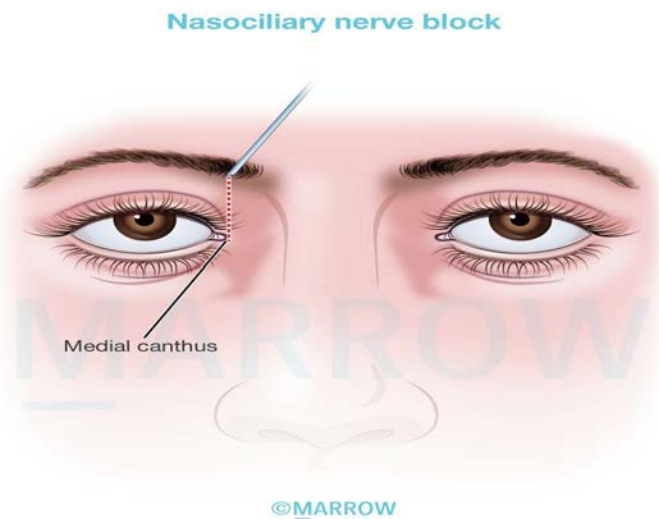
Branches:

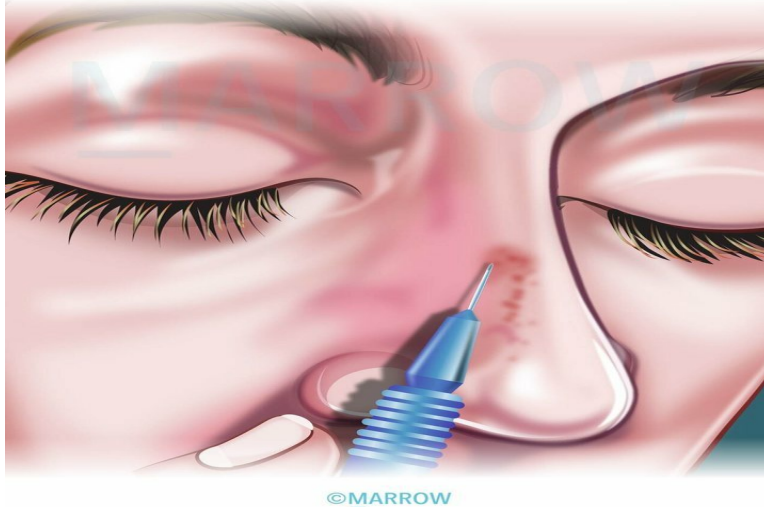
- Long ciliary nerve- cornea, ciliary body, iris
- Ramus communicans to the ciliary ganglion
- Posterior ethmoidal nerve- sensory innervation to frontal sinus and posterior ethmoidal air cells
- Anterior ethmoidal nerve- sensory innervation to middle and anterior ethmoidal cells as well as lateral part of the nasal septum
- Infra-trochlear nerve (continuation of the nasociliary nerve)- provides innervation to the conjunctiva, lacrimal sac, caruncle, the medial part of the eyelids

Uses of nasociliary nerve block:

Used in

- Surgery of nasal fractures,
- Septoplasty,
- Rhinoplasty and
- Repair of skin laceration.





The above image represents an anterior ethmoidal nerve block.

Solution to Question 4:

A patient, who has recovered from COVID-19 and hadn't required any hospitalization should be deferred from any elective surgery for a period of 6 weeks.

The American Society of Anesthesiologists has suggested the following duration of intervals for patients recovered from COVID-19. The wait time is from the date of diagnosis of COVID-19 to the date of surgery as follows:

- 4 weeks: for an asymptomatic patient or recovery from only mild, non-respiratory symptoms.
- 6 weeks: for a symptomatic patient (e.g: cough, dyspnea) who did not require hospitalization
- 8-10 weeks: for a symptomatic patient who is hospitalized or who has comorbidities such as diabetes or who is immunocompromised.
- 12 weeks: for a patient who was admitted in intensive care unit due to COVID-19 infection.

Solution to Question 5:

This patient has persistent hypoxia after an adequate trial of standard oxygen therapy and should be started on high-flow nasal cannula oxygen therapy.

When respiratory distress and/or hypoxemia of the patient cannot be alleviated after receiving standard oxygen therapy, high-flow nasal cannula oxygen therapy (HFNO), or non-invasive ventilation can be considered. Compared to standard oxygen therapy, HFNO reduces the need for intubation.

Patients on HFNO should be closely monitored, and endotracheal intubation should be done if the patient acutely deteriorates or does not improve after a short trial (about 1 hr).

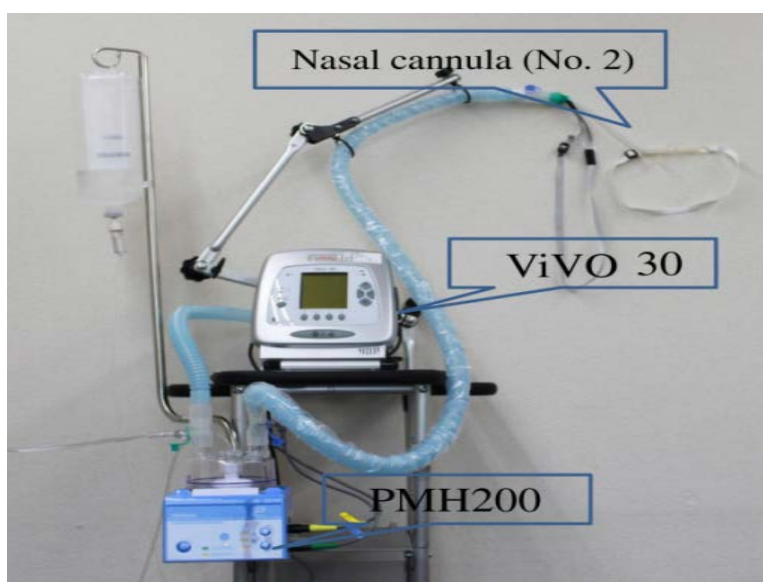
Patients with hypercapnia (exacerbation of COPD), hemodynamic instability, multi-organ failure, or abnormal mental status should not receive HFNO.

Note: If a high flow nasal cannula (HFNC) or simple nasal cannula is used to administer O₂, an N95 mask should be applied over it.

Some of the benefits of HFNC are

- It can deliver oxygen at 40L/min, 37°C, and 99.9% relative humidity.
- The high-flow oxygen is delivered directly to the nasopharynx. This produces FiO₂ equal to that delivered by traditional mask devices.
- It also reduces the work of breathing, improves gas exchange by nasopharyngeal dead-space washout, and decreases energy required to humidify and heat respiratory gases.

The HFNC equipment shown in the image below uses a heated humidifier (named PMH200), a nasal cannula (no. 2), and a ventilator (brand: vivo30).



Solution to Question 6:

Local anesthetics primarily act by blocking the sodium channels bound to the membrane.

Sodium channels have one larger α subunit and one or two smaller β subunits. These voltage-gated sodium channels exist in 3 states :

- Resting (non conducting)
- Open (conducting) and
- Inactivated (non conducting)

The local anesthetics bind to the α subunit of sodium channels, thereby preventing their activation and inhibiting the influx of sodium.

Local anesthetics have a higher affinity for the sodium channels in open and inactivated state than in the resting state.

Local anesthetic inhibition is both voltage and frequency-dependent and is higher when nerve fibers are firing rapidly than with slow, infrequent depolarizations.

Note:

Local anesthetics may also bind and inhibit calcium, potassium, transient receptor potential vanilloid 1 (TRPV1) and many other channels and receptors. But blocking of sodium channels is the best answer, of all the given options.

Solution to Question 7:

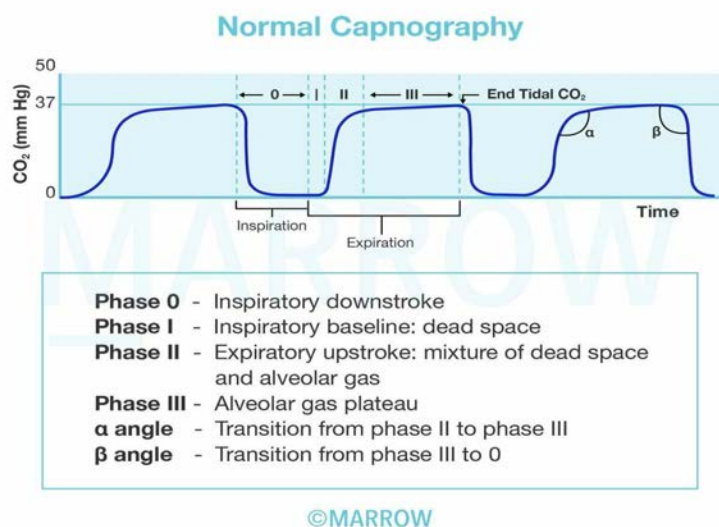
The reading marked as X in the monitor is the capnography.

Waveform capnography is a continuous non-invasive measurement and graphical display of the EtCO₂.

It is a direct measure of lung ventilation and an indirect measure of circulation and metabolism.

The normal capnograph displays the exhaled CO₂ in both inspiratory and expiratory phases. The expiratory phase is divided into three distinct components:

- Phase I: exhalation of dead space gas (i.e., partial pressure of CO₂, pCO₂ ~0).
- Phase II: A sharp rise in pCO₂ to a plateau indicates the sampling of transitional gas between the airways and alveoli.
- Phase III: The plateau region of the capnograph, corresponds to the pCO₂ in the alveolar compartment.
- Following phase III, a sharp inspiratory downstroke occurs, which is referred to as the beginning of phase 0 or phase IV by some.



Solution to Question 8:

Postoperative nausea and vomiting (PONV) is the most common cause of unanticipated hospital admission following daycare surgery.

The two most common complications of daycare surgery are PONV and postoperative pain. PONV refers to the presentation of symptoms in the postoperative care unit or within 24 hours of the procedure. Pain is usually managed preoperatively and postoperatively with the administration of analgesics. While it is a common complication, postoperative pain is not the most common cause of unanticipated hospital admission. Opioid medications which are commonly used to treat postoperative pain present with side effects such as nausea and vomiting which may aggravate PONV.

The occurrence of PONV can be based on the following factors, patient, choice of anesthetic agent, and the procedure. The risk factors are:

- Patient: Women (higher incidence in pregnancy), history of PONV or motion sickness, nonsmokers, younger age (incidence decreases as the age increases).
- Anesthetic agent: General anesthesia, inhalational agents (isoflurane, sevoflurane), nitrous oxide, longer duration of anesthesia, preoperative opioid administration.
- Procedure: Laparoscopic procedures, lithotripsy, gynecologic procedures, adenotonsillectomy, strabismus surgery.

The drugs used in the prophylaxis and treatment of PONV include ondansetron, dexamethasone, and transdermal scopolamine.

Solution to Question 9:

The commonly used frequency of tetanic stimulation in neuromuscular monitoring is 50 Hz.

Tetanic stimulation is a method to evaluate residual neuromuscular blockade.

Tetanic stimulation consists of a series of extremely rapid electrical stimuli typically applied over 5 seconds. The most commonly used frequency is 50 Hz stimulation given for 5 seconds.

- During a depolarizing block, the muscle response to 50 Hz tetanic stimulation for 5 seconds is sustained.
- In contrast, during a phase II block after the injection of succinylcholine, the response following tetanic stimulation is not sustained i.e., a fade occurs.

Disadvantages of tetanic stimulation are:

- This method has only 70% sensitivity to detect residual paralysis and its specificity is only about 50%. Hence, this test is of limited value for assessing neuromuscular recovery.
- This method is very painful, which makes it difficult to use in unanesthetized patients.
- During the late phase of neuromuscular recovery, tetanic stimulation can produce lasting antagonism of neuromuscular block in the stimulated muscle. This may change the response of the tested site such that it may no longer be representative of other muscle groups.

Note:

Other patterns of nerve stimulations, which are used to assess the patients for residual blockade are

- Train of four stimulation: Here four supramaximal stimuli (2 Hz) are delivered every 0.5 seconds.
- Double-burst stimulation: Here two short bursts of 50 Hz tetanic stimuli is delivered, which are separated by 750 ms.

Solution to Question 10:

A sudden fall in EtCO₂ with hypotension (fall in BP to 80/50 mmHg) along with normal airway pressure (< 35 to 40 cm of H₂O) during surgery is suggestive of CO₂ embolism.

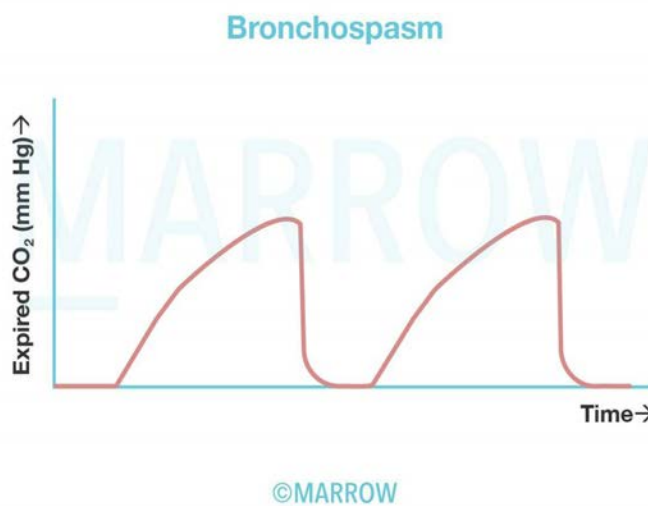
Intraoperative pulmonary embolism usually presents as sudden cardiovascular collapse and hypoxemia with a sudden fall in EtCO₂. Transesophageal echocardiography (TEE) is the most sensitive device for detecting intracardiac air embolus. If air is identified in the right atrium, or if it is suspected, immediate central vein cannulation and aspiration of the air may be lifesaving.

The placement of a vena cava filter should be considered postoperatively.

The best method to confirm the position of the endotracheal tube is the end-tidal CO₂ concentration assessed by capnography.

Other options:

Intraoperative bronchospasm is usually manifested as wheezing, increasing peak airway pressure, and a slowly rising waveform on the capnograph.



A pneumothorax may manifest as hypercapnia along with hypoxemia, hypotension, increased peak airway pressure, and decreasing tidal volumes. It may result in abrupt cardiovascular collapse unresponsive to fluid and vasopressor administration.

Deep sulcus sign on a supine chest radiograph points towards pneumothorax. The costophrenic angle is abnormally deepened when the pleural air collects laterally, producing the deep sulcus

sign.

The below chest X-ray shows left pneumothorax. Notice the collapsed lung and absent lung markings on the left side. Also mediastinal shift towards the right side.



Endobronchial intubation shows increased peak expiratory pressure, hypoxemia, increased airway pressure, asymmetric chest expansion and the absence of breath sounds over one lung, generally the left.

Note: Peak airway pressure is the amount of airway pressure delivered by the ventilator to overcome the resistance of the endotracheal tube, airway, and alveoli.

Solution to Question 11:

The maximum oxygen flow rate which can be kept with a conventional oxygen mask is 10 L/min.

A conventional oxygen mask is a simple face mask that is used routinely in hospitals. It covers the nose and mouth of the patient. There are 2 exhalation ports through which the patient exhales carbon dioxide. This mask is held in place with elastic around the back of the head and a metal piece that fits over the nose. It provides 40% to 60% of O₂ concentration and delivers O₂ at the rate of 6 to 10 L/min.

Hudson mask



©Marrow

Conventional oxygen therapy includes all the devices that are used in routine practices. It includes nasal prongs, low-flow nasal cannula, or masks. It is the main supportive treatment administered to patients after planned extubation.

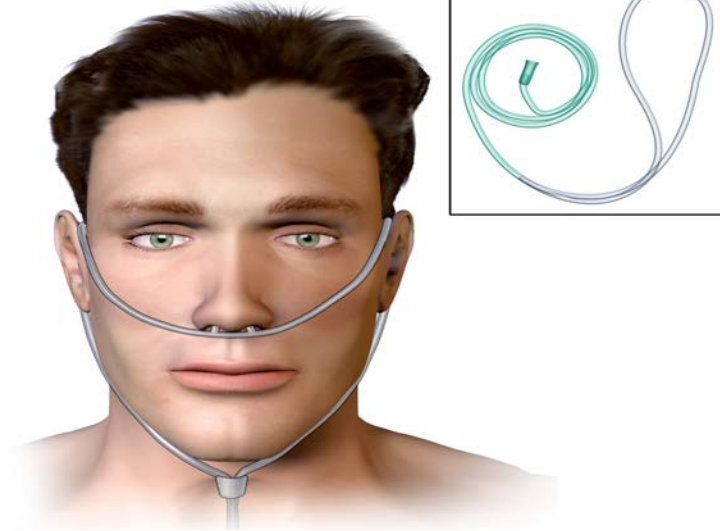
Oxygen delivery devices can be broadly divided into low flow oxygen devices (eg: nasal cannula, simple face mask, non-rebreathing mask) and high flow oxygen devices (eg: venturi mask).

Important note- If the question is about the maximum flow rate with the conventional oxygen mask, it would be 10L/ min. Whereas if the question is asking about the maximum oxygen flow rate with conventional oxygen therapy, then it would include all the conventionally used devices. Therefore, the answer would be 15L/min.

The below image shows a nasal cannula, which delivers oxygen at 1-6 L/min.

Oxygen delivery devices	Flow rate (liter/min)	Approximate F _i O ₂
Nasal cannula	1-6	0.24-0.44
Simple face mask	6-10	0.40-0.60
Non-rebreathing mask	10-15	0.90-1.00
Venturi mask	2-15	0.24-0.60

Nasal Cannula



The below image shows a non-rebreathing mask that provides an oxygen flow rate of up to 10-15 L/min.



The below image shows a venturi mask, which delivers oxygen at 2-15 L/min.

Venturi masks



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Solution to Question 12:

Among the given options, sevoflurane can be used for induction and maintenance of anesthesia. It can also be safely used in endotracheal intubation

Inhalation agents such as sevoflurane and halothane are useful in the induction of anesthesia, But sevoflurane due to its nonpungency and rapid onset makes it an excellent choice for smooth and rapid inhalation inductions in pediatric and adult patients. Sevoflurane has low blood solubility and partition coefficients resulting in more rapid emergence.

Sevoflurane also produces adequate muscle relaxation for intubation of children following an inhalation induction.

Other options:

Option A: Halothane can also be used for gaseous induction of anesthesia. Its disadvantage is long induction time owing to the high blood:gas partition coefficient. However, the incidence of halothane hepatitis and myocardial depression has reduced its use.

Option B: Desflurane produces laryngospasm when used for inhalational anesthesia and hence it is not used for induction. Laryngospasm is because of the pungent nature of the gas.

Option D: Isoflurane causes respiratory depression as well as upper respiratory irritation.

Solution to Question 13:

Patients who are unresponsive due to local anesthesia toxicity require 20% Intralipid injection during resuscitation.

Accidental intravascular injection of local anesthetic like bupivacaine results in bupivacaine-induced cardiotoxicity.

Initial management includes immediate initiation of cardiopulmonary resuscitation (CPR) and defibrillation as per the advanced cardiac life support (ACLS) algorithm.

- Following CPR, Intralipid therapy is initiated. A rapid bolus of Intralipid 20% IV (1.5 mL/kg) is administered without delay, followed by an infusion of 0.25 mL/kg/min. The infusion should be given continuously for at least 10 minutes after hemodynamic stability is achieved.
- Intralipid decreases cardiac bupivacaine concentration by 11% within 3 minutes of administration, and cerebral bupivacaine concentration by 18% within 15 minutes.

Bupivacaine is a long-acting local anesthetic that is the most cardiotoxic among the local anesthetics. It depresses the rapid phase of depolarization in Purkinje fibers and ventricular muscle due to a decreased availability of fast sodium channels in cardiac membranes. This reduced rate of depolarization depresses myocardial contractility. Hence, bupivacaine is a potent cardio-depressant.

Note: Local anesthesia systemic toxicity (LAST) can also affect the central nervous system (CNS), which is more susceptible to local anesthetic action as compared to the cardiovascular system. Dizziness, tinnitus, and disorientation are the initial symptoms. Shivering, muscular twitching, and tremors follow. Tonic-clonic seizures occur. Finally, respiratory depression can lead to respiratory arrest.

Solution to Question 14:

$FEV_1 < 0.8$ and $FEV_1/FVC < 0.7$ are classified as moderate COPD under GOLD classification and they require further evaluation of the lung function.

Preoperative pulmonary function tests (PFTs) assess the severity of known pulmonary disease, diagnose causes of respiratory symptoms, and may help to identify patients who are at risk for postoperative pulmonary complications.

The severity of COPD is assessed by using the Global Initiative for Lung Disease (GOLD) criteria. In pulmonary function testing, a postbronchodilator FEV_1/FVC ratio of < 0.70 is considered diagnostic for COPD.

The Global Initiative for Chronic Obstructive Lung Disease (GOLD) system categorizes airflow limitation into stages. In patients with $FEV_1/FVC < 0.70$:

- GOLD 1: mild: $FEV_1 \geq 80\%$ predicted
- GOLD 2: moderate: $50\% \leq FEV_1 < 80\%$ predicted
- GOLD 3: severe: $30\% \leq FEV_1 < 50\%$ predicted
- GOLD 4: very severe: $FEV_1 < 30\%$ predicted.

This is important in the perioperative period as the diagnosis of COPD is an independent and significant risk factor in the development of postoperative pulmonary complications.

Pre-operative FEV_1 and DLCO of $\geq 80\%$ imply a lower risk for a permanent postoperative pulmonary disability. Below these values, further risk-benefit assessment of the surgery needs to be examined and more information should be obtained.

Anesthesia NEET 2021

Question 1:

A patient undergoing surgery is maintained on anesthesia with halothane. During the procedure, the patient suddenly developed hyperthermia and muscle rigidity. Which of the following drugs is most likely implicated in this condition?

- a) D-curare
- b) Suxamethonium
- c) Cis-atracurium
- d) Rocuronium

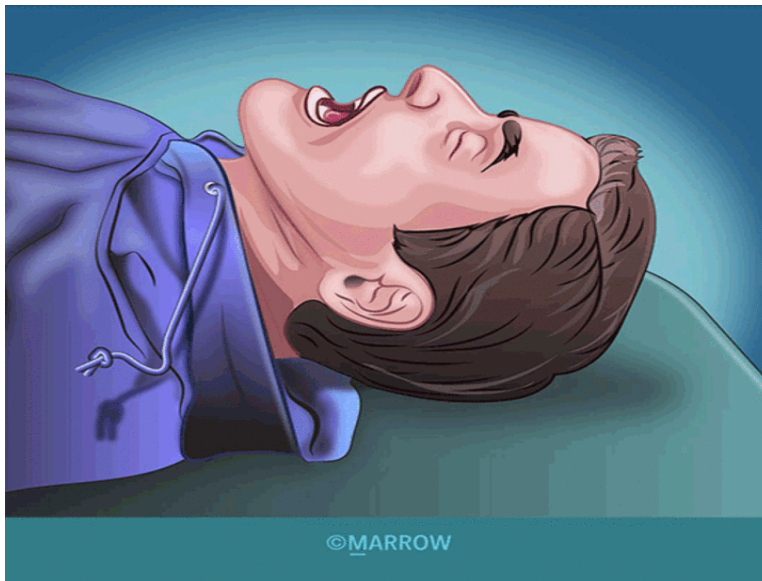
Question 2:

A 30-year-old male patient was intubated for surgery by a final-year resident. Which of the following is the best method to confirm the position of the endotracheal tube?

- a) X-ray chest
- b) Auscultation
- c) End-tidal CO₂ concentration
- d) Chest rise

Question 3:

What is the maneuver being performed in the given image?



- a) Head tilt, chin lift
- b) Jaw thrust
- c) Head extension
- d) In line manual stabilisation

Question 4:

A patient presented with headache, vomiting, and fever. Meningitis was suspected. While performing a lumbar puncture, which is the last structure you will pierce just before entering the lumbar cistern?

- a) Ligamentum flavum
- b) Dura mater
- c) Arachnoid membrane
- d) Pia mater

Question 5:

A young male patient was given regional block with 0.25% bupivacaine for a minor surgical procedure. Shortly after the injection, the patient suddenly became unresponsive and the pulse was not recordable. Which of the following would be the best course of management for this patient?

- a) CPR with 20% intralipid
- b) CPR with sodium bicarbonate

- c) CPR with dobutamine
- d) CPR with calcium

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	c
5	a

Detailed Explanations

Solution to Question 1:

The given clinical scenario of a patient being operated under halothane, developing hyperthermia and muscle rigidity is suggestive of an episode of malignant hyperthermia. The drug most likely implicating/causing this condition is suxamethonium (succinylcholine).

Suxamethonium is a peripherally acting depolarizing muscle relaxant, commonly administered during the induction of anesthesia to relax muscles of the jaw, neck, and airway and thereby facilitate laryngoscopy and endotracheal intubation.

Malignant hyperthermia is a rare genetic hypermetabolic muscle disease. It occurs due to mutation of the gene for the ryanodine receptor (Ryr 1), located on chromosome 19. Ryr 1 is an ion channel responsible for calcium release from the sarcoplasmic reticulum.

The symptoms develop when the patient is exposed to inhaled anesthetics (halothane in this case) or suxamethonium/succinylcholine.

The mutated Ryr 1 leads to a sudden release of calcium from the sarcoplasmic reticulum of the skeletal muscle. This uncontrolled increase in intracellular calcium removes the inhibition of troponin, resulting in sustained muscle contraction, which in turn, leads to an increase in adenosine triphosphatase (ATPase) activity. This results in an uncontrolled hypermetabolic state with increased oxygen consumption and CO2 production, producing severe lactic acidosis and hyperthermia.

The key clinical features are:

- Succinylcholine-induced masseter muscle rigidity (causing trismus) or generalized muscle rigidity
- Tachycardia

- Hypercarbia (increased CO₂ production) and reduced oxygen saturation
- Hyperthermia (rise of 1° Celsius can occur every 5 minutes)
- Arrhythmia
- Hypertension
- Myoglobinuria and myoglobinemia indicating muscle damage
- Metabolic derangements like hyperkalemia and mixed metabolic and respiratory acidosis

Treatment is to discontinue the triggering agents i.e; inhaled anesthetics and succinylcholine followed by administration of 100% oxygen. Metabolic acidosis and hyperkalemia need to be corrected. Hyperthermia must be controlled by cooling the patient.

Dantrolene sodium is the drug of choice. It is given at a dose of 2.5 mg/ kg intravenously and can be repeated every 5-10 minutes till the attack subsides.

Other options:

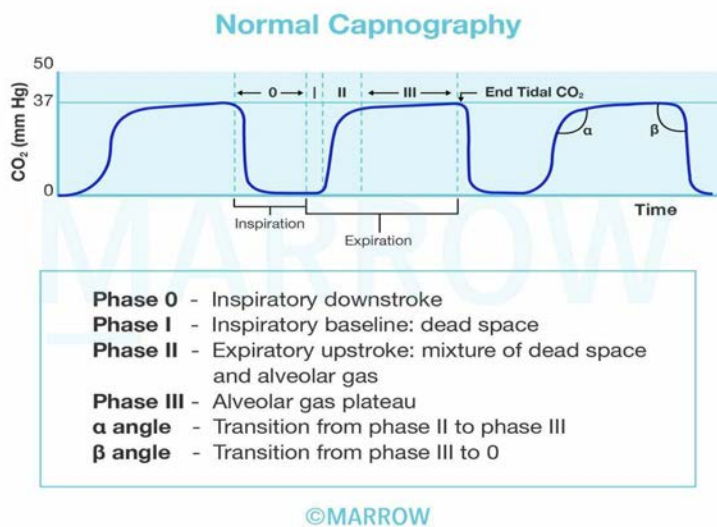
D-curare or D-tubocurarine (option A), cis-atracurium (option C), and rocuronium (option D) are non-depolarizing muscle relaxants and are not implicated in malignant hyperthermia.

Solution to Question 2:

The best method to confirm the position of the endotracheal tube is the end-tidal CO₂ concentration assessed by capnography.

Capnography is a non-invasive method of measurement and graphical display of partial pressure of carbon dioxide (CO₂) in exhaled breath or the end-tidal CO₂. Capnograph is the device used for measurement and capnogram is the CO₂ waveform that is displayed. It is the most reliable indicator of endotracheal tube placement.

The proper placement of the endotracheal tube is indicated by a normal CO₂ waveform for at least three breaths.



Other signs to confirm proper endotracheal intubation are:

Symmetrical chest rise, presence of equal bilateral breath sounds, absent breath sounds over the epigastrium, visible condensation in the ETT, large exhaled tidal volume, and appropriate compliance of the reservoir bag during manual ventilation.

Absent CO₂ waveform tracing despite correct ETT intubation is seen in the following conditions:

- Severe bronchospasm,
- Equipment malfunction,
- Cardiac arrest, or hemodynamic collapse

Clinical relevance:

- Capnography does not differentiate endotracheal from endobronchial intubation. Signs of endobronchial intubation include increased peak expiratory pressure, hypoxemia, increased airway pressures, asymmetric chest expansion and the absence of breath sounds over one lung, generally the left.
- In esophageal intubation, the capnograph can initially detect some CO₂ in the stomach from swallowing expired air. But after a few breaths, this CO₂ will be washed out and ETCO₂ will become zero.
- The most sensitive and earliest indicator of malignant hyperthermia is a persistent increase in ETCO₂.

Solution to Question 3:

The maneuver being performed in the given image is that of head-tilt chin-lift. It is an important technique in emergency airway management to open up the airway.

Basic life support involves early recognition and management of emergencies such as heart attack, stroke, and foreign body airway obstruction. The sequence of BLS is circulation (chest compression), airway, and then breathing.

The airway is usually obstructed by posterior displacement of the tongue or epiglottis. One hand (palm) is placed on the patient's forehead applying pressure to tilt the head back. Simultaneously, the chin is lifted with the forefinger and index finger of the opposite hand. This causes cervical flexion with an atlantooccipital extension (sniffing position). This lifts the tongue from the back of the throat.

Head-tilt chin-lift should not be performed in a patient with suspected cervical injury.

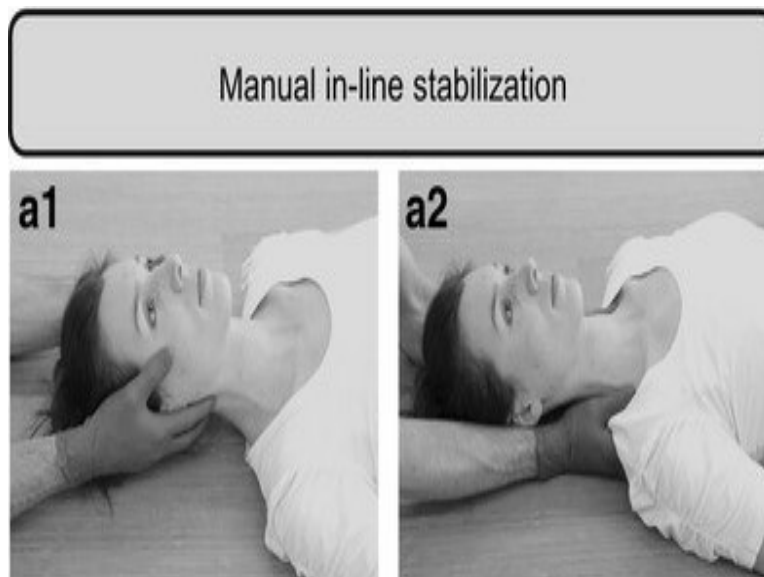
Other options:

Option B: Jaw thrust- Both hands of the clinician are placed on either side of the patient's head and the angles of the jaw are grasped and lifted. It can be performed in a patient with cervical injury but caution must be followed to not hyperextend the neck.



Jaw Thrust

Option D: Manual inline stabilization is a maneuver that limits external head and neck movement in patients with cervical spine injuries. Fingers and palms of both hands are used to stabilize the patient's occiput and mastoid processes to ensure successful airway intervention.



Solution to Question 4:

The arachnoid membrane is the last structure you will pierce while performing a lumbar puncture, just before entering the lumbar cistern.

Lumbar puncture is a diagnostic procedure done to obtain a sample of cerebrospinal fluid from the subarachnoid space. During the procedure, the following structures are pierced to reach the subarachnoid space, in the given order:

- Skin

- Subcutaneous tissue
- Supraspinous ligament
- Interspinous ligament
- Ligamentum flavum
- Dura mater
- Arachnoid membrane

Indications:

- Diagnosis of suspected meningitis and subarachnoid hemorrhage
- Spinal anesthesia and intrathecal administration of antibiotics/chemotherapy
- Therapeutic relief of pseudotumor cerebri, less frequently

Contraindications for lumbar puncture include:

- Thrombocytopenia or other bleeding diatheses (including anticoagulant therapy)
- Raised intracranial pressure
- Spinal epidural abscess
- Infection over the site

Site of the procedure:

In adults, the spinal cord ends at the level of the lower border of the L1 vertebra. Lumbar puncture is usually performed at L3-L4 space (below L1 vertebra) to avoid potential needle trauma to the spinal cord.

Technique:

The patient is usually placed in the lateral recumbent position. An alternative position is the sitting position, useful in obese patients to determine the midline. L3-L4 space is located by palpating the right and left posterior superior iliac crests and moving the fingers medially toward the spine at the same level.

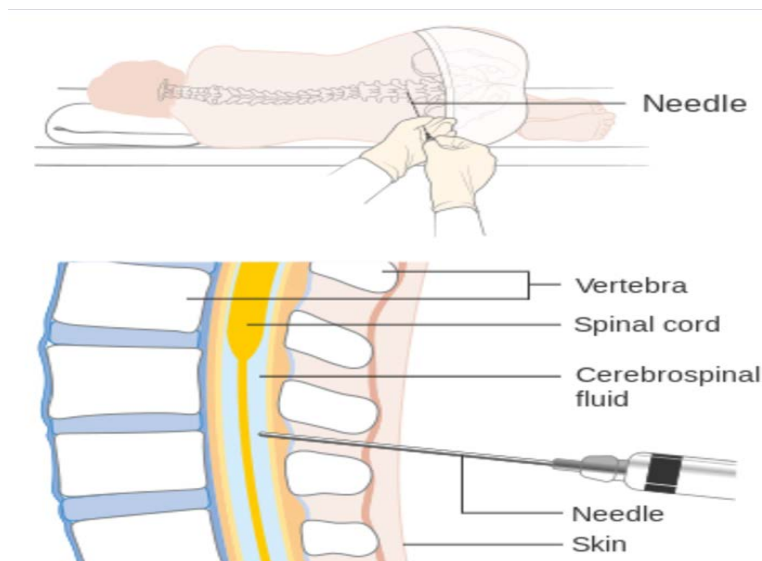
A local anesthetic like 1% lidocaine is administered. A 22-gauge needle is advanced through the skin. In the lateral recumbent position, the bevel should face up, and in the sitting position, it should face to one side or the other, to avoid injury to the dural fibers. There is a loss of resistance as the needle enters the epidural space. Dura and then arachnoid are next pierced, to release CSF.

Complications following lumbar puncture:

Post-dural puncture headache (PDPH) is a common post-procedure symptom. It occurs due to loss of CSF through the dura which causes traction on the pain-sensitive intracranial structures. The headache in PDPH is typically frontal or occipital. The smaller the needle used, the lesser the risk of PDPH.

Other complications include infections, bleeding, cerebral herniation, and radicular pain.

The image below shows a patient undergoing lumbar puncture in the lateral recumbent position:



Solution to Question 5:

The clinical picture of an unresponsive patient with an absent pulse following bupivacaine administration is most likely due to bupivacaine-induced cardiotoxicity resulting from accidental intravascular injection of local anesthetic. The best course of management in this scenario would be administering CPR (cardiopulmonary resuscitation), along with a 20% intralipid injection.

Bupivacaine is a long-acting local anesthetic that is the most cardiotoxic among the local anesthetics. It depresses the rapid phase of depolarization in Purkinje fibers and ventricular muscle due to a decreased availability of fast sodium channels in cardiac membranes. This reduced rate of depolarization depresses myocardial contractility. Hence, bupivacaine is a potent cardio depressant.

Cardio-vascular effects of bupivacaine-induced toxicity are as follows:

- Left ventricular depression
- Atrioventricular heart block
- Ventricular arrhythmias and ventricular fibrillation
- Sinus bradycardia can lead to cardiac arrest

Initial management includes immediate initiation of cardiopulmonary resuscitation (CPR) and defibrillation as per the advanced cardiac life support (ACLS) algorithm.

Following CPR, intralipid therapy is initiated. Its mechanism is explained by the lipid sink theory which states lipophilic drugs (bupivacaine) are absorbed into the lipid emulsion of the plasma, hence, removed from tissues affected by toxicity.

A rapid bolus of intralipid 20% IV (1.5 mL/kg) is administered without delay, followed by an infusion of 0.25 mL/kg/min. The infusion should be given continuously for at least 10 minutes after hemodynamic stability is achieved.

Local anesthesia systemic toxicity (LAST) can also affect the central nervous system (CNS), which is more susceptible to local anesthetic action as compared to the cardiovascular system. Dizziness, tinnitus, and disorientation are the initial symptoms. Shivering, muscular twitching, and tremors follow. Tonic-clonic seizures occur. Finally, respiratory depression can lead to respiratory arrest.

Anesthesia INI-CET 2022

Question 1:

Arrange the following anesthetic agents on the basis of their potency.

- a) Methoxyflurane> Halothane> Isoflurane> Nitrous oxide
- b) Methoxyflurane> Isoflurane> Halothane> Nitrous oxide
- c) Methoxyflurane> Nitrous oxide> Halothane> Isoflurane
- d) Halothane> Isoflurane> Nitrous oxide> Methoxyflurane

Question 2:

Which of the following local anaesthetic causes hypertension?

- a) Cocaine
- b) Lignocaine
- c) Bupivacaine
- d) Prilocaine

Question 3:

Which of the following is required for the cannulation of a newborn?

- a) 1,3,4,5
- b) 2,3,4,5
- c) 2,4
- d) 1,4

Question 4:

Identify the incorrect statement in rapid sequence intubation?

- a) Preoxygenation with 100% oxygen for 3 min is done.
- b) High-volume low-pressure cuff is used.
- c) It is the method of choice in cardiac arrest patient.

d) Induction agent and muscle relaxant are given simultaneously.

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	c

Detailed Explanations

Solution to Question 1:

The potency of the inhaled anesthetics given above can be arranged as methoxyflurane > halothane > isoflurane > nitrous oxide. It is due to the change in the agents' minimum alveolar concentration (MAC).

The minimum alveolar concentration (MAC) of an inhaled anesthetic is the lowest concentration required to prevent movement in response to a defined noxious stimulus in 50% of subjects. It is a measure of the potency of inhalational anesthetics. So, the lower the MAC value, the lower the concentration of anesthetic drug required, implying the drug is more potent.

MAC values (expressed as a '%' of 1 atmosphere) of some anesthetic agents:

- Nitrous oxide with a MAC value of 104 is the least potent inhaled anesthetic
- Methoxyflurane – MAC- 0.2 (most potent)
- Desflurane: 6.0
- Sevoflurane– 2.05
- Isoflurane – 1.28
- Halothane – 0.75

Solution to Question 2:

The local anesthetic that causes hypertension is cocaine. This is due to the inhibition of norepinephrine at the adrenergic terminals leading to increased sympathetic stimulation. This effect occurs at all concentrations of cocaine and causes hypertension and ventricular ectopies.

Treatment is with the help of benzodiazepines, α -adrenergic antagonists and amiodarone.

All local anesthetics are mainly cardiodepressants causing reduced contractility and vasodilation. However, at low concentrations, all local anesthetics cause vasoconstriction due to the inhibition of nitric oxide. Only cocaine produces vasoconstriction and hypertension at all concentrations.

Accidental intravascular injection of local anesthetic like bupivacaine results in bupivacaine-induced cardiotoxicity. Management is by resuscitation followed by intralipid therapy. A rapid bolus of intralipid 20% IV is administered without delay.

Other adverse effects of local anesthetics include neurologic toxicity, respiratory depression, and hypersensitivity reactions. Nervous system manifestations are circumoral numbness, tongue paresthesia, dizziness, tinnitus, and disorientation. Finally, respiratory depression can lead to respiratory arrest.

Solution to Question 3:

For cannulation in newborns, we would require a 24G needle, 5 mL syringe, adhesive tape, and normal saline.

Also, for a peripherally inserted central catheter (PICC), we use a 24G needle for ≥ 1 kg neonate, and for < 1 kg neonate, we use a 28G needle. We need to avoid using 1 mL syringes for flushing as it may lead to catheter rupture; preferably, use 5 or 10 mL syringes. Flushing of the cannulation is done using 0.9% sodium chloride or so-called normal saline (NS).

Dressing change: We change PICC line dressing only when visibly soiled, damp, or loosened. For making it easier and sterile, two persons should perform a dressing change.

Solution to Question 4:

The incorrect statement regarding rapid sequence intubation (RSI) is that it is the method of choice in cardiac arrest patients.

RSI is a technique of anesthesia induction in patients who are at an increased risk of gastric regurgitation and pulmonary aspiration of gastric contents. Common cases include pregnancy, full stomach, intestinal obstruction, poorly controlled gastroesophageal reflux disease, morbid obesity, and diabetic gastroparesis. The goal is to intubate rapidly to minimize the time between the loss of consciousness and securing of the airway with a cuffed endotracheal tube.

It is achieved by pre-oxygenating the patient with 100% oxygen for 2-3 minutes. An inducing agent is administered, which is immediately followed by a muscle relaxant succinylcholine or rocuronium. The trachea is intubated without any prior bag and mask ventilation using a high-volume and low-pressure cuff which adequately cuffs the airway and prevents aspiration.

Cardiac arrest is an emergency condition with importance given to start chest compressions, defibrillation, or drugs to restore perfusion. The airway is commonly blocked by posterior displacement of the tongue which can be managed by head tilt and chin lift or giving jaw thrust. In case of airway blockade or to prevent aspiration, the airway is maintained by emergency tracheal tube intubation and does not involve the procedure that is followed in RSI. Endotracheal tube placement is commonly done during post-cardiac arrest care.

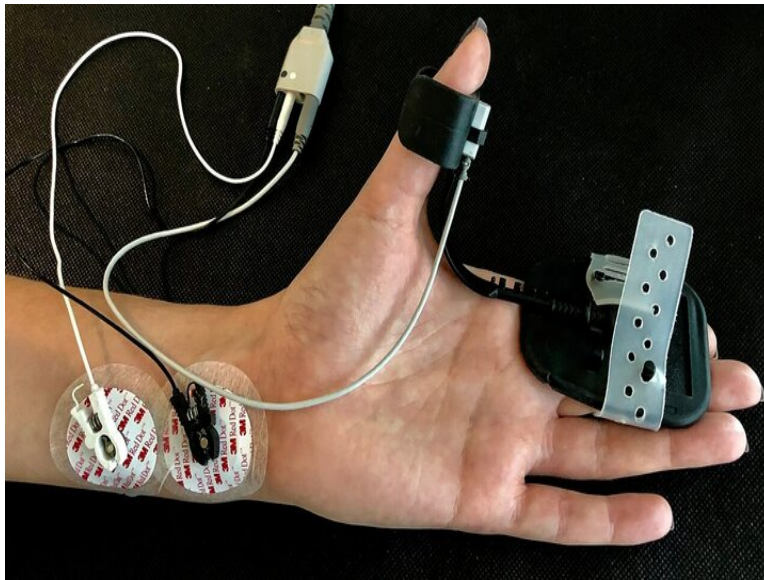
RAPID SEQUENCE INDUCTION



Anesthesia NEET 2022

Question 1:

The image given below shows neuromuscular monitoring of the patient after anesthesia. What is the most commonly used nerve for monitoring?



- a) Ulnar nerve
- b) Median nerve
- c) Radial nerve
- d) Metacarpal nerve

Question 2:

Identify the mask in the image used for patients with COVID-19 infection.



- a) Venturi mask
- b) Hudson mask
- c) Nebuliser
- d) Non-rebreathing mask

Answer Key

Question No.	Correct Option
1	a
2	d

Detailed Explanations

Solution to Question 1:

The following image shows neuromuscular monitoring and the most commonly used nerve for monitoring is the ulnar nerve.

The mechanosensor with electrodes is placed over the region of the ulnar nerve. This assesses the contraction of the adductor pollicis muscle to monitor neuromuscular activity.

Monitoring of neuromuscular blockade is required to monitor the blockade intraoperatively, ensure its complete reversal postoperatively, and prevent residual blockade. To monitor the neuromuscular function, a peripheral motor nerve is supramaximally stimulated and

the response of the muscle is assessed.

The most commonly assessed nerve-muscle unit is the ulnar nerve-adductor pollicis muscle. This is because it is easily accessible, has the lowest risk of direct muscle stimulation, and the response can be evaluated visually, by tactile methods, or by objective methods. To stimulate the ulnar nerve, the electrodes are best applied to the volar aspect of the wrist.

Other nerve-muscle units that can be assessed include:

- Posterior tibial nerve- flexor hallucis brevis muscle
- Facial nerve- orbicularis oculi muscle
- Facial nerve- corrugator supercilii muscle

Many methods can be used for objective monitoring of neuromuscular function:

- Mechanomyography (MMG)
- Electromyography (EMG)
- Acceleration of the muscle response (AMG)
- Kinemyography (KMG)
- Cuff pressure modality (CPM)
- Compressomyography (CMG)
- Phonomyography (PMG)

Solution to Question 2:

The mask in the image is a non-rebreathing mask (NRBM). It can be used for patients with COVID-19 infection.

A NRBM has a reservoir bag attached to the mask in order to provide additional reservoir volume. In addition, there is a valve attached, which provides a unidirectional flow of gases. Flutter valves on the side ports prevent room air entrainment, while a valve incorporated between the mask and the reservoir bag prevents the entry of the exhaled gases.

NRBM can provide the highest FiO_2 of 0.9 to 1.0. without intubation at the fresh gas flow of 12-15 L/min. They are used for preoxygenation before intubation and post-extubation in covid patients.

Other options:

Option A: The venturi mask is a high-flow, fixed-performance device that delivers a specific oxygen concentration to the patient. It is a color-coded device that connects to the oxygen tubing distally and has a nozzle connecting to a mask proximally. Oxygen flows through a central constriction inside the device. Room air enters through surrounding apertures which dilutes the oxygen to the required concentration. The venturi mask is based on Bernoulli's principle.

Venturi masks



©Marrow

Option B: Hudson mask is a simple face mask that delivers oxygen without forming a tight seal against the patient's face. There are holes on either side of Hudson's mask. This provides air entrainment, meeting the peak inspiratory flow and venting out the expired gases.

Hudson mask



©Marrow

Option C: A nebulizer is used to deliver drugs to the breathing system. It generates an aerosol, producing droplets of varying sizes. It is shown below.



Anesthesia NEET 2023

Question 1:

Identify the maneuver being performed in the given image.



- a) Head tilt, chin lift
- b) Jaw thrust
- c) Head extension
- d) In line manual stabilization

Question 2:

Which of the following is the drug of choice for preoperative antibiotic prophylaxis in a patient undergoing cardiac surgery?

- a) Penicillin G
- b) Erythromycin
- c) Azithromycin
- d) Cefazolin

Question 3:

During resuscitation, when is the given position indicated?



- a) Unconsciousness with pulse and breathing absent
- b) Unconsciousness with pulse present and breathing absent
- c) Unconsciousness with pulse and breathing present
- d) Unconsciousness with pulse absent and breathing present

Answer Key

Question No.	Correct Option
1	a
2	d
3	c

Detailed Explanations

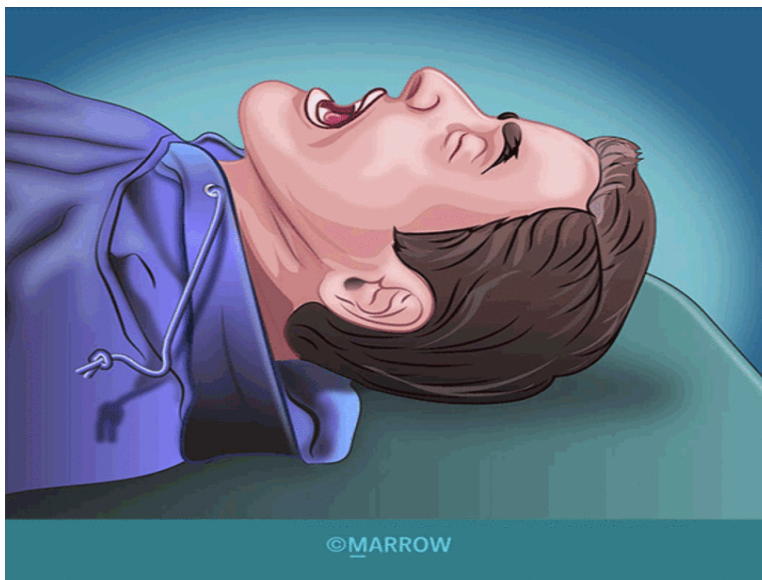
Solution to Question 1:

The maneuver performed in the given image is that of a head-tilt chin-lift. It is an important technique in emergency airway management to open up the airway.

Basic life support (BLS) involves early recognition and management of emergencies such as heart attack, stroke, and foreign body airway obstruction. The sequence of BLS is circulation (chest compression), airway, and breathing.

The airway is usually obstructed by posterior displacement of the tongue or epiglottis. One hand (palm) is placed on the patient's forehead, applying pressure to tilt the head back. Simultaneously, the chin is lifted with the forefinger and index finger of the opposite hand. This causes cervical flexion with an atlantooccipital extension (sniffing position). This lifts the tongue from the back of the throat.

Head-tilt chin-lift should not be performed in a patient with suspected cervical injury.



Other options:

Option B: Jaw thrust - Both hands of the clinician are placed on either side of the patient's head, and the angles of the jaw are grasped and lifted. It can be performed in a patient with cervical injury, but caution must be exercised to not hyperextend the neck.



Jaw Thrust

Option D: Manual inline stabilization is a maneuver that limits external head and neck movement in patients with cervical spine injuries. Fingers and palms of both hands are used to stabilize the patient's occiput and mastoid processes to ensure successful airway intervention.



Solution to Question 2:

First- and second-generation cephalosporins (cefazolin) are the drugs of choice for preoperative prophylaxis for cardiac surgery. Clindamycin is reserved for cases of allergy to beta-lactams and vancomycin is recommended for methicillin-resistant *Staphylococcus aureus* (MRSA).

Preoperative antibiotic prophylaxis is essential in reducing the risk of surgical site infections, particularly in cardiac surgery, where the consequences of infection can be severe. The choice of antibiotic depends on the patient's medical history, allergies, and the most common pathogens involved in surgical site infections for the specific surgery. In addition to cefazolin, other cephalosporins, such as cefuroxime, can also be used for prophylaxis in cardiac surgery. The timing of antibiotic administration is crucial, with the first dose typically given within 60 minutes before the incision.

Other Options:

Option A: Penicillin G is not the drug of choice for preoperative antibiotic prophylaxis in cardiac surgery. It is mainly used to treat infections caused by susceptible gram-positive bacteria, such as *Streptococcus pyogenes*, *Streptococcus pneumoniae*, and some strains of *Staphylococcus aureus*.

Option B: Erythromycin is a macrolide antibiotic and is not the drug of choice for preoperative prophylaxis in cardiac surgery. It is typically used to treat respiratory tract infections, skin infections, and other infections caused by susceptible gram-positive bacteria.

Option C: Azithromycin, another macrolide antibiotic, is also not the drug of choice for preoperative prophylaxis in cardiac surgery. It is commonly used to treat respiratory tract infections, sexually transmitted infections, and other infections caused by susceptible gram-positive and gram-negative bacteria.

Solution to Question 3:

The position illustrated in the picture is known as the recovery position, and it is used in the management of patients who are unconscious but with pulse present and breathing present.

The recovery position can help prevent complications such as aspiration and airway obstruction in unconscious individuals. Additionally, it is crucial to monitor the patient's airway, breathing, and circulation continuously while they are in the recovery position and seek immediate medical assistance. An unresponsive patient who is breathing spontaneously in the supine position is at risk of airway obstruction from their tongue, mucus, vomit, etc. A modified lateral position is used and the following 6 principles are considered while choosing which variation of recovery position to use,

-

The victim should be in as near a true lateral position as possible, with the head dependent to allow free drainage of fluid.

-

The position should be stable.

-

Avoid any pressure on the chest that impairs breathing.

-

It should be possible to turn the victim on his or her side and to return to the back easily and safely, with concern for a possible cervical spine injury.

-

Good observation of and access to the airway should be possible.

-

The position itself should not cause injury to the victim.

Other options

Option A: Unconsciousness with pulse and breathing absent - In this situation, the patient is in cardiac arrest and requires immediate cardiopulmonary resuscitation (CPR), not the recovery position. CPR involves chest compressions and rescue breaths to support circulation and respiration.

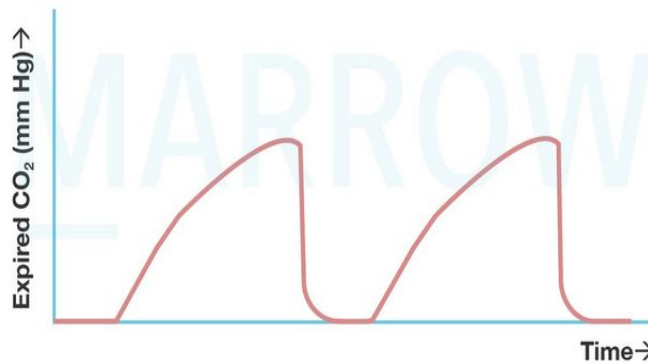
Option B: Unconsciousness with pulse present and breathing absent - In this case, the patient is experiencing respiratory arrest. Rescue breathing or bag-valve-mask ventilation should be initiated to support the patient's breathing. The recovery position is not indicated in this scenario.

Option D: Unconsciousness with pulse absent and breathing present - This option is not physiologically plausible, as the absence of a pulse implies a lack of circulation, and breathing cannot be present without circulation.

Anesthesia INI-CET 2023

Question 1:

A patient who is under general anaesthesia shows the following finding on the ET CO₂ monitor. Which condition can lead to this characteristic waveform on capnography?



- a) Malignant hyperthermia
- b) Bronchospasm
- c) Rebreathing
- d) Accidental extubation

Question 2:

In a patient with thrombocytopenia, platelet transfusion is done when the platelet level drops below a threshold. To proceed with surgery, platelet transfusion is done to achieve a platelet level of?

- a) 30,000
- b) 40,000
- c) 50,000
- d) 60,000

Question 3:

The presence of which of the following indicates that CPR is of high quality?

- a) 2, 3 only
- b) 1, 3, 4 only
- c) 1, 2, 3, 4
- d) 1 only

Question 4:

A 70 kg patient who presented to the emergency with acute pancreatitis was shifted to the ICU and intubated. His ventilatory settings are: FiO₂ 90%, PEEP 12 mmHg, respiratory rate 16/min, and tidal volume 420 ml. The results of the ABG analysis are given below. Which of the following would be the next step in the management of this patient?

- a) Increase PEEP to 16
- b) Increase tidal volume
- c) Reduce respiratory rate
- d) Reduce FiO₂

Question 5:

Which of the following best represents the American Society of Anesthesiologists (ASA) physical status III?

- a) Mild systemic disease
- b) Moderate systemic disease
- c) Severe systemic disease
- d) Severe systemic disease with threat to life

Question 6:

A 5-year-old is on 3L/min O₂ via a nasal cannula. What is the appropriate FiO₂ level for this child?

- a) 25
- b) 30

- c) 45
- d) 60

Answer Key

Question No.	Correct Option
1	b
2	c
3	b
4	d
5	c
6	b

Detailed Explanations

Solution to Question 1:

The above capnographic waveform depicts bronchospasm.

Shark fin appearance - Prolonged phase II, increased α angle, and steeper phase III suggest bronchospasm or airway obstruction.

Solution to Question 2:

In an otherwise healthy patient, surgery can safely be performed in patients with platelet counts higher than 50,000/mm3. The risk of bleeding increases significantly if the platelet count is below this level. When platelet transfusions are used to treat thrombocytopenia, the platelet count rises by 10,000/mm3 for every unit transfused.

Thrombocytopenia is defined as a platelet count less than 1,50,000/mm3. It may be the result of immune thrombocytopenic purpura (ITP), drug-induced thrombocytopenia, rheumatological diseases (SLE, rheumatoid arthritis), pregnancy and chronic liver disease. The initial step if the patient has low platelet count is to repeat the test and examine the peripheral smear.

Anemia is also a common preoperative hematological disorder. The optimal preoperative hemoglobin concentration varies between 7.5 to 10 g/dL across the individuals, with inter-individual differences due to co-morbid conditions in the patients. Patients with anemia or suspected anemia must undergo a battery of investigations to determine the type of anemia and how best to correct it. Preexisting anemia is a risk factor for postoperative death and

complications such as AKI, stroke and infections.

Solution to Question 3:

Chest compressions at a rate of 100-120/minute, depth of compression 5-6 cm, and allowing complete chest recoil indicate high-quality CPR. The compression-to-ventilation ratio in the absence of an advanced airway should be 30:2 (30 compressions to be followed by 2 breaths). If an advanced airway (endotracheal tube or supraglottic airway device) is present, 10 breaths/minute should be given.

Solution to Question 4:

In a patient with a history of acute pancreatitis who needed to be intubated, the diagnosis of acute respiratory distress syndrome (ARDS) is the most likely. In this patient, the PaO₂ is 110 mmHg and the SpO₂ is 100%, which are significantly higher than the target recommended in patients with ARDS. The next step in the management of this patient would be to reduce the FiO₂ to prevent any further lung injury and worsening of ARDS due to oxygen toxicity. In the initial phase of the management, the FiO₂ of the patient should be kept $\leq 60\%$ and SpO₂ should be maintained between 88-95%.

The main treatment strategy which has been shown to reduce mortality in patients is low tidal volume ventilation (6 mL/kg of predicted body weight). Increasing the tidal volume (option B) in this patient will lead to barotrauma to the alveoli because air is pushed through the alveoli which are already collapsed.

Although a high PEEP keeps the alveoli opened up, there is no rationale for increasing the PEEP further (option A) in this patient because the oxygenation is adequate. Furthermore, PEEP must be adjusted to minimize FiO₂ and provide adequate PaO₂ at the same time.

Hypercapnia is expected in a patient on low tidal volume ventilation. It can be managed by increasing the respiratory rate. However, respiratory acidosis is tolerated by the patient up to a blood pH of 7.2. This is known as permissive hypercapnia. Reducing the respiratory rate (option C) in this patient will increase the PaCO₂ further and cause a worsening of respiratory acidosis.

Apart from these, prone position, early neuromuscular blockade (to prevent the patient from fighting the ventilator), and use of extracorporeal membrane oxygenation (ECMO) have also been shown to improve mortality in patients with ARDS. Management strategies that have not been shown to improve the outcome in ARDS patients include glucocorticoids, surfactant replacement, anti-inflammatory therapies like NSAIDs, and inhaled vasodilators.

Solution to Question 5:

Severe systemic disease best fits into grade 3 of the ASA classification.

American Society of Anesthesiologists classification is used to define relative risk prior to conscious sedation and surgical anesthesia.

It is simple, reproducible and strongly correlates to perioperative risk.

American Society of Anesthesiologists physical status classification of patients-

Class 1: Normal healthy patient

Class 2: Patient with mild systemic disease (no functional limitations)

Class 3: Patient with severe systemic disease (some functional limitations)

Class 4: Patient with severe systemic disease that is a constant threat to life (functionality incapacitated)

Class 5: A moribund patient who is not expected to survive without the operation

Class 6: A brain-dead patient whose organs are being removed for donor purposes

Solution to Question 6:

The appropriate FiO₂ for this child would be 30.

The formula for calculating FiO₂ for a child on oxygen supplementation through nasal cannula is
$$FiO_2 = 21\% + (\text{nasal cannula flow in L/min}) \times 4$$

A nasal cannula is a low flow device, and FiO₂ should be adjusted according to flow rates.

Examples of low-flow devices are:

- Nasal cannula
- Face mask
- Face tent
- Non-rebreather mask.

High-flow devices include:

- Venturi mask
- High flow nasal cannula
- Mechanical ventilation

Oxygen delivery devices	Flow rate (liter/min)	Approximate FiO ₂
Nasal cannula	1-6	0.24-0.44
Simple face mask	6-10	0.40-0.60
Non-rebreathing mask	10-15	0.90-1.00
Venturi mask	2-15	0.24-0.60

Anesthesia INI-CET 2024

Question 1:

Which of the following is a 2nd generation LMA device?

- a) Intubating LMA
- b) Proseal LMA
- c) Flexible LMA
- d) Ambu LMA

Question 2:

A patient was planned for surgery under general anaesthesia. During induction of anaesthesia, rocuronium 85 mg was given and the patient was tried to be intubated. It resulted in a failed intubation. Which of the following drugs can be given for immediate reversal of the drug?

- a) Neostigmine
- b) Glycopyrrolate
- c) Sugammadex
- d) Edrophonium

Question 3:

Which of the following are components of high-quality cardio-pulmonary resuscitation(CPR) in adults?

- a) 1,2 and 3
- b) 1,3 and 4
- c) 2,3 and 4
- d) 3 and 4 only

Question 4:

What is the inhalational anaesthetic agent used for induction in children?

- a) Nitrous oxide
- b) Isoflurane
- c) Sevoflurane
- d) Desflurane

Question 5:

Ambu Aura 40 is an example of?

- a) Supraglottic laryngeal airway generation 1
- b) Supraglottic laryngeal airway generation 2
- c) Supraglottic laryngeal airway generation 3
- d) Supraglottic laryngeal airway generation 4

Question 6:

Which of the following drugs causes adrenal suppression?

- a) Thiopental
- b) Ketamine
- c) Etomidate
- d) Propofol

Question 7:

Which of the following is incorrect regarding thiopentone?

- a) It is lipid soluble
- b) It achieves maximum concentration in the brain after an IV infusion
- c) Redistribution to adipose tissue and muscle is seen
- d) Rapid decline in its effect is due to renal excretion

Question 8:

A female following an RTA injured her chest. On X-ray, a fracture of 3 to 5th ribs was noted. Pain was not responsive to analgesics. Which block will be given to reduce the pain?

- a) Cervical block
- b) Thoracic epidural block
- c) Supraclavicular block
- d) Lumbar epidural block

Question 9:

Which of the following will lead to a decrease in diastolic BP?

- a) Systemic infusion of alpha-adrenergic blocker
- b) Systemic infusion of smooth muscle relaxant
- c) Loss of lumbar sympathetic outflow
- d) Systemic infusion of beta-1 adrenergic agonist

Question 10:

Which of the following is true regarding paediatric cardiopulmonary resuscitation?

- a) Chest compression at the rate of 30:2 with 2 rescuers
- b) Adenosine via interosseous is preferred over other methods
- c) In chest compression, depth must be at least 1/3rd of the child's chest AP diameter
- d) Adrenaline given in dose 0.01 mg/kg iv/io in 1:1000 concentration

Question 11:

Which of the following must be avoided in post-cardiac arrest resuscitation care?

- a) Maintain MAP over 65
- b) Maintain SpO₂ above 94%
- c) Revascularisation procedure if the patient develops ST elevation post-resuscitation
- d) Bring the body temperature down to below 32 degrees, therapeutic hypothermia if unresponsive

Question 12:

Jackson Rees circuit is used in

- a) Paediatric anaesthesia
- b) Cardiac anaesthesia
- c) Neuro anaesthesia
- d) Obstetric anaesthesia

Answer Key

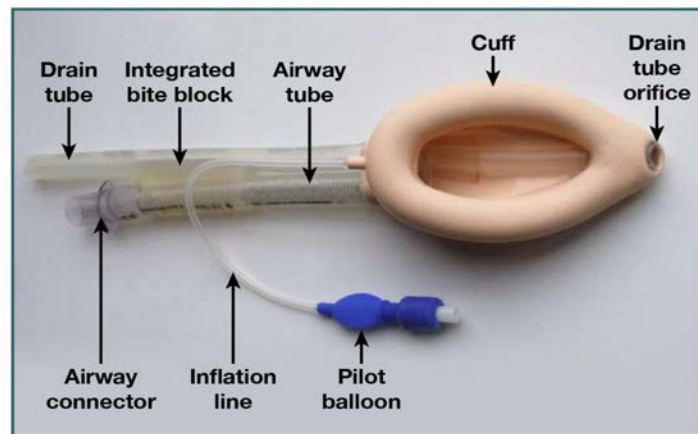
Question No.	Correct Option
1	b
2	c
3	b
4	c
5	a
6	c
7	d
8	b
9	c
10	c
11	d
12	a

Detailed Explanations

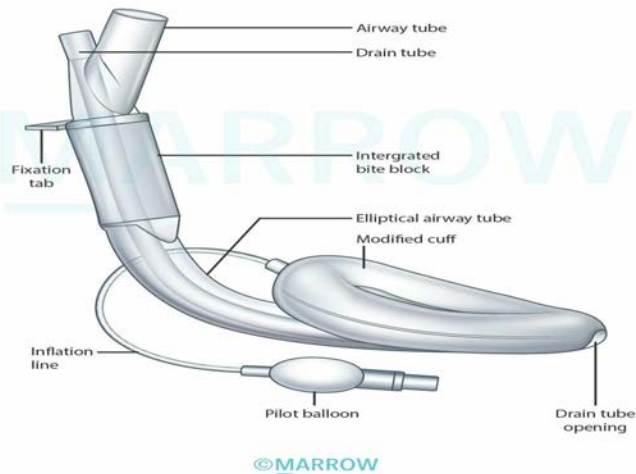
Solution to Question 1:

Proseal LMA is a 2nd generation LMA device as it has a separate gastric tube to decompress the stomach and prevent gastric aspiration associated with the classical LMA or 1st generation LMA. Other 2nd generation LMA's include LMA Supreme, I-Gel and LMA Protector.

Proseal Laryngeal Mask Airway



Laryngeal Mask Airway Supreme



All other options are examples of 1st generation LMA's since they do not have a separate gastric tube to aspirate for gastric secretions.

Flexible LMA allows the tube complete flexibility and resistance to compression so that the head and neck can be turned without dislodging the mask.

The intubating LMA (ILMA) and Ambu LMA(ALMA) are variants of the LMA Classic, a 1st generation LMA used to aid in intubation of the patient in difficult airways.



Note : Although Ambu LMAs are produced as 1st generation and 2nd generation LMA, Proseal is exclusively a 2nd generation LMA and hence the best answer to this question.

Solution to Question 2:

Sugammadex is used for immediate reversal of rocuronium induced neuromuscular blockade/relaxant in the given scenario.(Option C)

Sugammadex is a modified γ -cyclodextrin and the first selective relaxant-binding agent based on an encapsulating principle for inactivation of nondepolarising neuromuscular blocking drugs mainly rocuronium and vecuronium. This principle of complex formation of sugammadex and rocuronium or vecuronium in a 1:1 ratio occurs at all levels of neuromuscular blockade resulting in a more fast-acting pharmacologic reversal when compared with anticholinesterase drugs like neostigmine.

The stability of the rocuronium-sugammadex complex is a result of Van der Waals forces, hydrogen bonds and hydrophobic interactions. There is some binding affinity with pancuronium, but this interaction is too low to have a significant clinical effect.

Other options:

Option A: Neostigmine is used to reverse depolarising neuromuscular blockade such as succinylcholine, especially in the phase II block(fade of the train-of-four response) and sufficient time has passed for the circulating concentration of succinylcholine to be negligible. Therefore, neostigmine is slow-acting compared to sugammadex but the agent preferred in the case of succinylcholine.

Neostigmine administration during phase I block, increases neuromuscular Ach concentration and inhibits pseudocholinesterase (plasma cholinesterase) activity that can prolong the action of succinylcholine, hence it is given after some time during the phase II block.

Option B: To prevent cholinergic side-effects of neostigmine, glycopyrrolate is co-administered in a 1:5 ratio to neostigmine(0.2mg: 1 mg). Atropine is an alternative but glycopyrrolate is preferred

due to the incidence of tachycardia associated with atropine.

Option D: Edrophonium is a short-acting acetylcholinesterase inhibitor preferred in the case of mivacurium reversal since it does not show plasma cholinesterase inhibition present in neostigmine that can prolong mivacurium action.

Solution to Question 3:

The depth of compression should be at least 5cm to ensure high-quality CPR in an adult cardiac arrest.

The foundation of CPR is high-quality chest compressions. Compressing the chest pumps blood from the heart to the brain and then to the rest of the body. Each time the compressions are stopped, the blood flow from the heart decreases significantly. Once resumed, it takes several compressions to bring the blood flow back up to the levels present before the interruption.

The components of high-quality chest compressions in adults are as follows :

- Position victim - Position the victim face up on a firm, flat surface. This allows compression of the heart to create adequate blood flow.
- Compression-to-ventilation ratio - Single rescuers should use the compression-to-ventilation ratio of 30 compressions to 2 breaths when giving CPR to victims of any age.
- Compression rate - Compress at a rate of 100 to 120/min. This rate is the same for compression in all cardiac arrest victims.
- Compression depth - Compress the chest at least 5cm. Compressing the chest more than 6 cm in adults may decrease the effectiveness of the compression and cause injuries. Using a CPR-quality feedback device can help reach the optimal depth.
- Chest recoil - Allow the chest to recoil completely after each compression. This allows blood to flow into the heart. To help ensure complete recoil, avoid leaning on the chest between compressions. Chest compression and chest recoil times should be about equal.
- Interruptions in chest compressions - Minimize interruptions in chest compression. the proportion of time that rescuers perform chest compressions during CPR is called chest compression fraction (CCF). A CCF of 80% or higher should be the goal.

Solution to Question 4:

Sevoflurane is the inhalational anaesthetic agent used for induction in children. (Option C)

It is the preferred agent due to its non-pungent sweet-smelling nature and rapid increase in alveolar concentration on administration. Its low blood solubility helps during emergence from the effects of anaesthesia, reducing the post-operative delay in the patient's recovery from anaesthesia.

Desflurane and isoflurane are avoided for inhalation induction because they are pungent and associated with coughing, breath-holding and laryngospasm. They are usually used for the maintenance of anaesthesia. Desflurane is preferred in patients with renal and liver disease and

isoflurane is preferred in neurosurgery and cardiac surgery.

Due to its second-gas effect, nitrous oxide is mainly used as a carrier gas for other inhalational agents and not given alone.

Solution to Question 5:

The Ambu Aura 40 is an example of a first-generation supraglottic airway device.

Supraglottic airway (SGA) devices are classified into different generations based on their features and advancements:

First-generation SGA devices: Basic devices designed to provide a seal around the glottis for ventilation. They lack additional safety features, such as protection against aspiration. Examples include the original Laryngeal Mask Airway (LMA Classic) and Ambu Aura 40 (can be reused up to 40 times),

Second-generation SGA devices: These are improved devices with additional features to enhance safety and functionality. They typically have:

A gastric access channel (to allow gastric decompression or reduce the risk of regurgitation and aspiration).

Improved sealing mechanisms. Examples: Ambu Aura Gain (with drain tube), LMA ProSeal, and i-gel.

Third-generation and Fourth-generation devices: These are not standard classifications in clinical practice yet. However, devices beyond the second generation may feature advanced integration with monitoring systems, adjustable seals, or further innovations.

Solution to Question 6:

Etomidate causes adrenal suppression which can last for 72 hours.

Etomidate inhibits 11 β -hydroxylase, resulting in decreased biosynthesis of cortisol and mineralocorticoids and increases the formation of intermediaries like 11-deoxycorticosterone.

It is a more potent inhibitor of steroid synthesis than as a sedative-hypnotic as the concentration which produces adrenocortical suppression (10 ng/mL) is much lower than that needed for hypnosis (>200 ng/mL).

This suppression is particularly concerning in critically ill patients, where stress-induced cortisol production is vital for survival.

Other Options:

Option A: Thiopental: Primarily causes sedation and hypnosis.

Option B: Ketamine: Does not cause adrenal suppression. Instead, it may slightly stimulate adrenal catecholamine release due to its action on the sympathetic nervous system.

Option D: Propofol: Can cause CNS depression without direct impact on adrenal hormone synthesis.

Solution to Question 7:

The rapid decline in the effect of thiopentone is due to its rapid redistribution, not due to renal excretion.

Thiopentone has a rapid onset of action because of its lipophilic nature, so it rapidly crosses the blood-brain barrier and enters highly perfused organs like the brain, heart, liver, and kidneys.

As thiopentone redistributes from the brain to tissues: it perfuses to muscle, fat, and skin. Over time, due to its high lipid solubility, thiopentone accumulates maximally in fat. Fat acts as a storage reservoir since it has low blood flow, and drug release from fat occurs slowly.

This redistribution reduces the concentration of thiopentone, terminating its anesthetic effect.

Other Options:

Option A: It is lipid-soluble: Thiopentone is highly lipid soluble, which allows it to cross the blood-brain barrier quickly and achieve rapid onset of action in the CNS.

Option B: It achieves maximum concentration in the brain after an IV infusion: Due to its lipophilicity, thiopentone rapidly accumulates in the brain after IV administration.

Option C: Redistribution to adipose tissue and muscle is seen: Thiopentone's effect is terminated by redistribution from the brain to adipose tissue and muscle. This accounts for the short duration of its anesthetic effect.

Solution to Question 8:

A thoracic epidural block will be given to reduce the rib fracture pain.

Epidural anaesthesia is used to provide adequate analgesia which is the mainstay of treatment in patients with multiple rib fractures.

The complications can be minimized by adequate pain relief with epidural anaesthesia. Rib fracture patients will have the following complications due to splinting and pain:

- Hypoventilation
- Hypoxia

Other techniques to provide analgesia following rib fractures are intercostal nerve block and intrapleural block which necessitates multiple injections, which increases the risk for complications.

The complications include vascular puncture, pneumothorax, patient discomfort due to the need for palpation of the fractured rib during local anaesthetic (LA) administration. Also, LA toxicity can occur because of the high systemic absorption.

Rib fractures can cause severe pain that does not respond well to conventional analgesics. A thoracic epidural block is effective for pain relief in this area because it provides segmental anesthesia to the thoracic dermatomes and surrounding structures, helping reduce pain and improve respiratory function.

Indications of thoracic epidural block:

- Postoperative pain relief: After thoracic or upper abdominal surgeries.
- Trauma management: Rib fractures to reduce pain and improve ventilation.
- Chronic pain syndromes: Postherpetic neuralgia or cancer-related pain.

Other Options:

Option A: Cervical block: Used primarily for pain management or anesthesia involving the neck and upper shoulders.

Option C: Supraclavicular block: Used for anesthesia of the upper limb by blocking the brachial plexus.

Option D: Lumbar epidural block: Used for lower body anesthesia, including the pelvis and lower extremities.

Solution to Question 9:

Loss of lumbar sympathetic outflow will lead to a decrease in diastolic blood pressure (DBP).

Loss or blockade of the lumbar sympathetic chain (spinal anaesthesia) causes dilation of capacitance (venous) vessels. This leads to a decrease in diastolic blood pressure and subsequent orthostatic lightheadedness or syncope.

Vasodilation is more marked on the venous than on the arterial side of the circulation, resulting in blood pooling in the venous capacitance vessels. This reduction in circulating blood volume is well tolerated at low levels of spinal anaesthesia in healthy patients. With an increasing level of block, this effect becomes more marked, and venous return becomes gravity-dependent. If venous return decreases too much, cardiac output and organ perfusion decline sharply.

Other options:

Option A: Systemic infusion of alpha-adrenergic blocker will minimally decrease diastolic BP, not significantly like in loss of lumbar sympathetic outflow.

Option C: Systemic infusion of smooth muscle relaxants like Sildenafil, which is a PDE 5 inhibitor (increases cGMP), does not cause a decrease in DBP.

Option D: Systemic infusion of beta-1 adrenergic agonist increases systolic BP and heart rate.

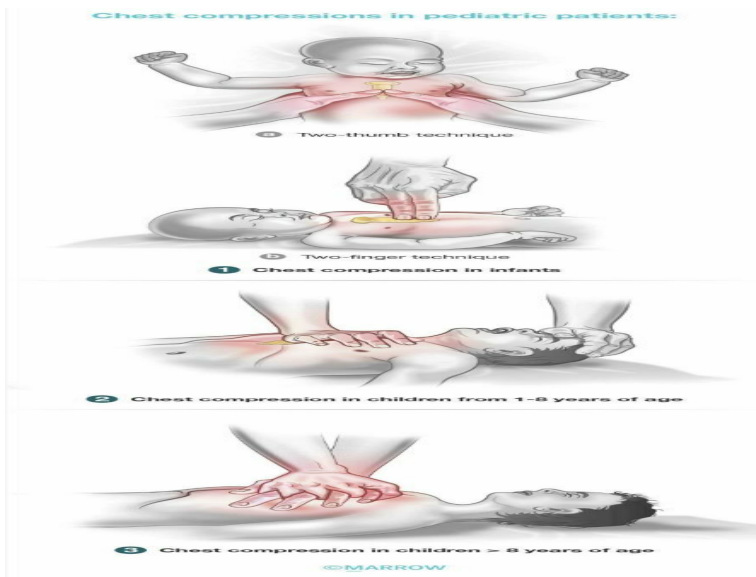
Solution to Question 10:

The depth of compressions in infants should be at least 1.5 inches or 1/3 the depth of the infant's chest.

Effective chest compressions in infants and pediatric population:

- 100 to 120 compressions per minute (30 compressions for every 15-18 sec).
- One provider: Place two fingers, one between the nipple line and the other 1 cm below the first finger

- Two providers: Encircle the infant's torso with both hands, thumbs pointing cephalad and positioned 1cm below the nipples over the sternum.
- Depth of compressions should be at least 1.5 inches or 1/3 the depth of the infant's chest.
- Press hard and fast
- Allow for full chest recoil after every compression



Other Options:

Option A: Chest compression at the rate of 30:2 with 2 rescuers: For children, the compression-to-ventilation ratio is 15:2 when there are two rescuers. A 30:2 ratio is used only if there is a single rescuer, similar to adult CPR.

Option B: Adenosine via interosseous is preferred over other methods: Adenosine is not used in pediatric cardiopulmonary resuscitation. It is used in PSVT treatment in intravenous routes.

Option D: Adrenaline given in dose 0.01 mg/kg iv/io in 1:1000 concentration: Epinephrine is the drug of choice for treating bradycardia in infants and children. The recommended dose is 0.01 mg/ kg of a 1:10,000 solution.

Solution to Question 11:

The body temperature in post cardiac arrest care is to be maintained between 32-36 degrees for at least 24 hours. This is known as targeted temperature management.

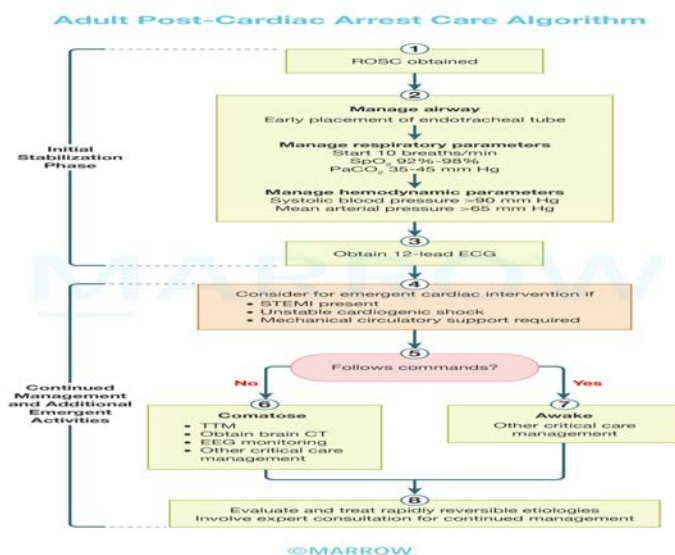
Targeted temperature management (TTM) is the only intervention demonstrated to improve neurological recovery after cardiac arrest and involves maintaining core body temperature between 32-36°C for at least 24 hours.

This is done in post-cardiac resuscitation patients who have achieved a return of spontaneous circulation (ROSC) but are still comatose.

The entire body is to be cooled using a combination of methods like fluid-filled cooling blankets, ice packs, forced air blankets, cold IV fluids, and/or invasive devices or catheters.

It is the only intervention that can improve neurological recovery after cardiac arrest. It is most beneficial in ventricular fibrillation and pulseless ventricular tachycardia patients.

Cardiac arrest (especially with shockable rhythms) and neonatal hypoxic-ischemic encephalopathy are the primary established indications for therapeutic hypothermia. Stroke (in some cases), where thrombectomy is not an option, is another indication.



Other Options:

Option A: Maintain MAP over 65: A MAP of ≥ 65 mmHg is the minimum threshold recommended during post-cardiac arrest care to prevent ischemic injury and support organ function.

Option B: Maintain SpO₂ above 94%: The recommended target SpO₂ range in post-cardiac arrest care is 92-98%.

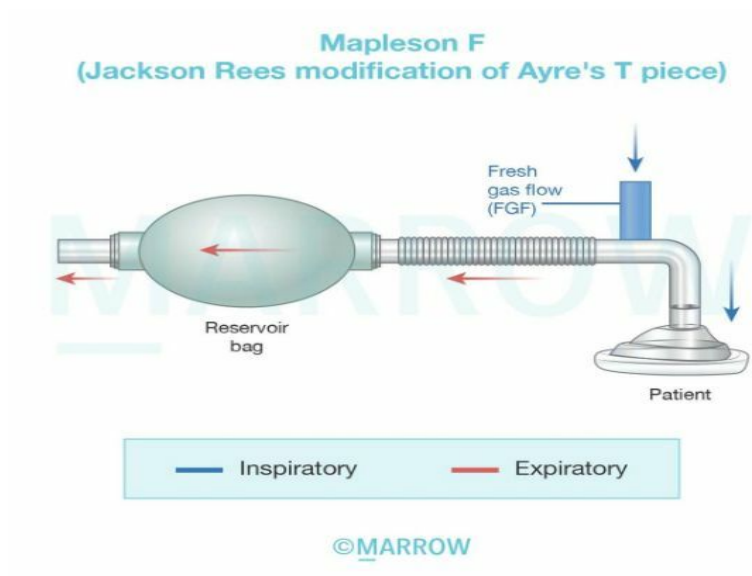
Option D: Revascularisation procedure if the patient develops ST elevation post resuscitation: Post-cardiac arrest patients with STEMI benefit from revascularization, as it reduces mortality and improves cardiac function.

Solution to Question 12:

The Jackson Rees circuit is used in paediatric anaesthesia.

Jackson Rees modification of Ayre's T piece is classified as a Mapleson's F system (Semi-Closed Circuits).

It is used in children < 6 years or < 20 kgs.



Other options:

Option B, C, and D: Cardiac anaesthesia, Neuro anaesthesia, and Obstetric anaesthesia require closed circuit. They have a provision to absorb CO₂, using soda lime. Closed circuits are used for long periods of procedures.

Mapleson Circuit	Usage
Type A (Magill's circuit)	Circuit of choice for spontaneous ventilation
Type B	Obsolete
Type C (Water's to and fro)	Obsolete
Type D (Bain's circuit)	Circuit of choice for controlled ventilation. Most commonly used circuit.
Type E (Ayre's T piece)	Circuit of 2nd choice in children
Type F (Jackson Rees Modification)	Circuit of choice in children (<20 kg)

Anesthesia NEET 2024

Question 1:

A patient undergoing surgery in the prone position is maintained on mechanical ventilation with PEEP. What is the significance of using PEEP in this scenario?

- a) Prevents alveoli collapse and atelectasis of lung
- b) Decreases inspiratory pressure
- c) Increases venous return and maintains cardiac output
- d) Reduces CO₂ in alveoli

Question 2:

Which of the following statements are true about eutectic mixture of local anesthetic (EMLA)?



- a) 1gm of 0.5% of cream contains 50mg lignocaine
- b) Methohemoglobinemia is a side effect
- c) It is an equal mixture
- d) It is made of Lignocaine and ropivacaine

Question 3:

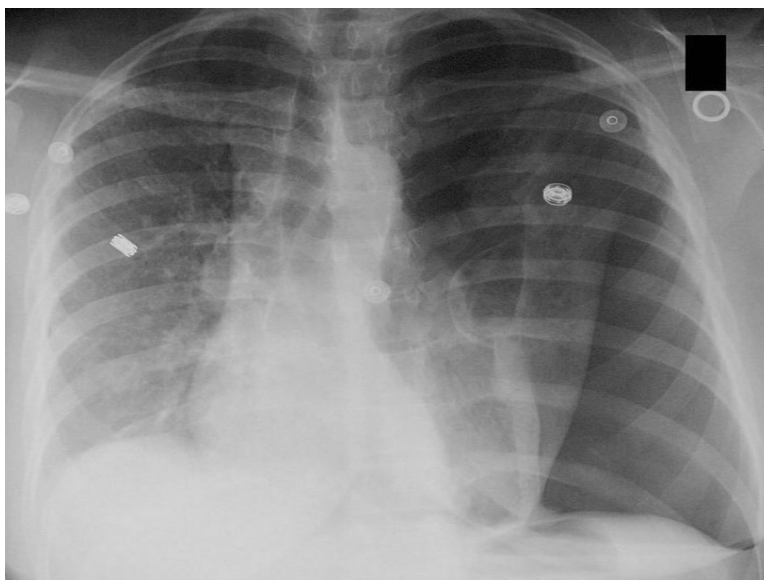
An anesthetist used the airway device shown in the image given below on a patient. Which among the following statements is true regarding the adjunct?



- a) It can cause damage to the teeth
- b) It can be used in conscious patients
- c) It does not cause laryngospasm
- d) It can be safely used in facial trauma cases

Question 4:

In the following condition as shown in the radiograph, which of the following procedures would be difficult to perform? "



- a) Nasogastric tube insertion
- b) Laryngeal mask airway
- c) Tracheostomy
- d) Indirect laryngoscopy

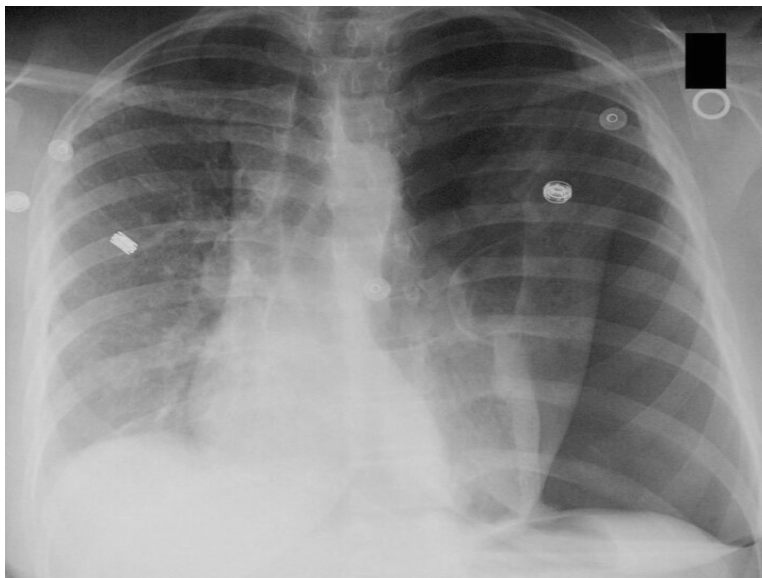
Question 5:

Which of the following agents is appropriate for reversing vecuronium when renal function tests are normal?

- a) Physostigmine
- b) Rivastigmine
- c) Sugammadex
- d) Succinyl choline

Question 6:

A COVID-19 patient admitted to the ICU on mechanical ventilation suddenly develops hypoxia. The patient has decreased breath sounds on the left side and increased intrathoracic pressure. What is the most likely diagnosis?

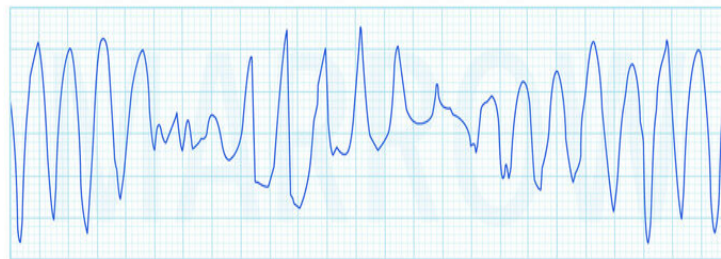


- a) Pneumothorax
- b) Left-Sided Pleural Effusion

- c) Acute Respiratory Distress Syndrome
- d) Left-Sided Pneumonia

Question 7:

A 3-year-old child presents with drowsiness and refusal to feed. On examination, the child is in a stuporous state. The ECG is shown below. What is the next best step in the management of this patient?

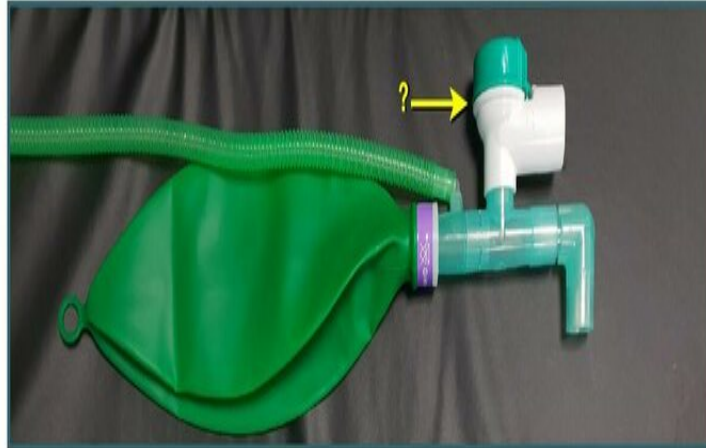


©Marrow

- a) Immediate synchronized cardioversion @ 1 joule/kg
- b) Defibrillate, check rhythm and then start CPR
- c) Defibrillate followed by CPR for 2min, and check rhythm
- d) Stop CPR and do cardioversion at 2 joules/ kg

Question 8:

At what pressure does the marked structure open?



- a) 30-40 cm H₂O
- b) 30-40 mm Hg
- c) 10-20 cm H₂O
- d) 10-20 mm Hg

Question 9:

This recovery position is appropriate for:

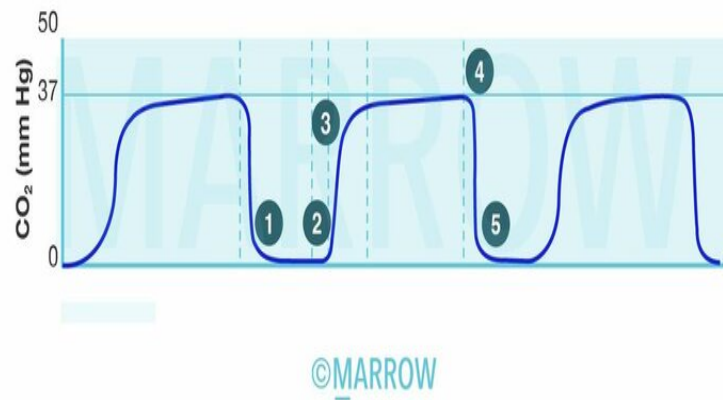


- a) Unresponsive patient, pulse and breathing absent
- b) Responsive patient, pulse and breathing present

- c) Unresponsive patient, pulse and breathing present
- d) Unresponsive with pulse, without breathing

Question 10:

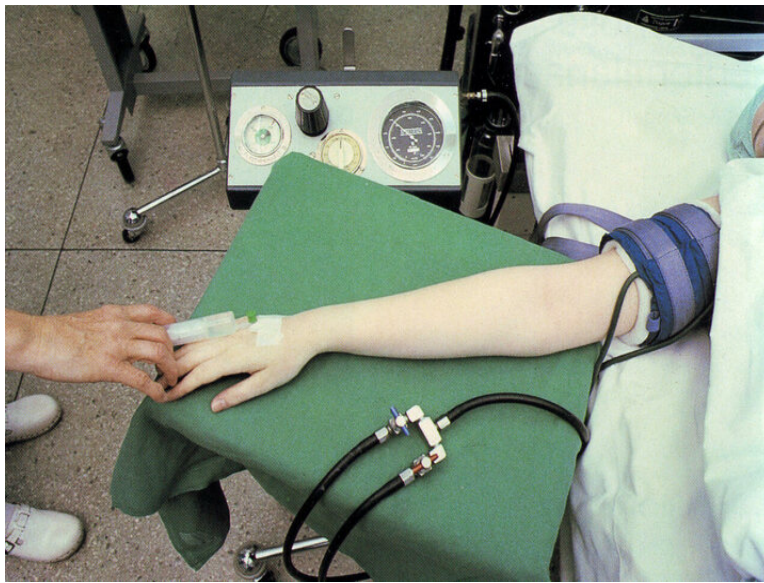
Following is the image of a normal capnograph. What is true about the segment 3-4?



- a) Inhalation of fresh gas
- b) Exhalation of dead space air
- c) Exhalation of mixed gases
- d) Alveolar gas and plateau pressure

Question 11:

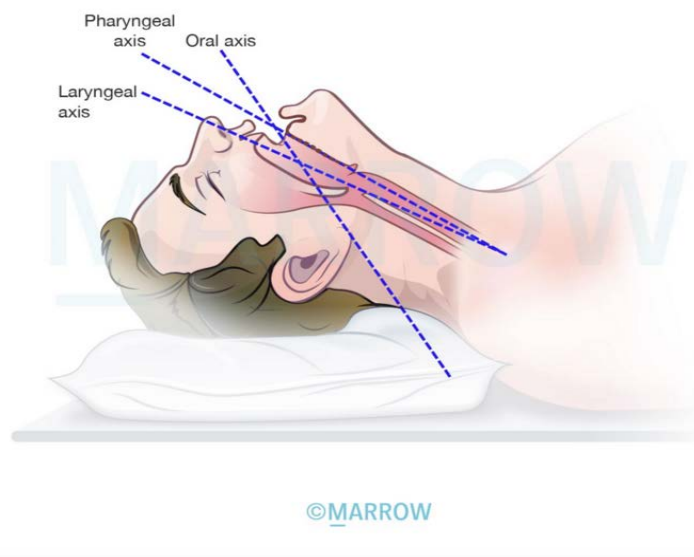
What is the procedure being done in the image?



- a) Bier Block
- b) Axillary nerve block
- c) Median nerve block
- d) Wrist block

Question 12:

What is true about the following position?



- a) High rate of error
- b) Ideal position for bougie insertion in emergencies

- c) Ideal position for neurosurgery
- d) Sniffing position for intubation

Question 13:

A patient presents with stridor after prolonged intubation. The endoscopic view is shown in the image below. What is the staging system for this condition?



- a) Cotton Meyer's
- b) AJCC
- c) Radkowski
- d) TNM

Question 14:

A patient is undergoing implant surgery for his upper limb. On giving general anaesthesia he develops tachycardia, sudden raise in body temperature and rigidity of muscles. What drug is most likely to have caused this?

- a) Propofol
- b) Etomidate
- c) Nitrous oxide
- d) Halothane

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	c
5	c
6	a
7	c
8	a
9	c
10	d
11	a
12	d
13	a
14	d

Detailed Explanations

Solution to Question 1:

The significance of using positive end-expiratory pressure (PEEP) in the given clinical scenario is to help keep the alveoli open, preventing collapse (atelectasis), which is especially important in the prone position where certain lung regions might be more susceptible to collapse.

This is crucial in conditions where the lungs are prone to collapse, such as Acute Respiratory Distress Syndrome (ARDS) or when the patient is in a prone position during surgery. By maintaining open alveoli, PEEP improves gas exchange, which increases oxygen levels in the blood.

Other options:

Option B: Decreases inspiratory pressure: PEEP can increase the overall inspiratory pressure by maintaining pressure in the lungs at the end of expiration.

Option C: Increases venous return and maintains cardiac output: PEEP can decrease venous return due to increased intrathoracic pressure, which may reduce cardiac output.

Option D: Reduces CO₂ in alveoli: PEEP does not directly reduce CO₂ in the alveoli. The effect on CO₂ depends on ventilation and perfusion matching, not just PEEP alone.

Solution to Question 2:

Methemoglobinemia can develop when high doses of prilocaine are administered, typically over 600 mg.

EMLA (Eutectic Mixture of Local Anaesthetic) is a topical mixture containing prilocaine. The risk of methemoglobinemia is increased in neonates with metabolic disorders or in those receiving drugs that can impair the reduction of methemoglobin.

It is an eutectic(in fixed proportions) mixture, not necessarily equal (Option C). It consists of a 1:1 mixture of 2.5% lidocaine (Option A) and 2.5% prilocaine, not ropivacaine (Option D)

EMLA can provide adequate dermal analgesia for procedures like:

- Split skin-graft harvesting
- Venipuncture
- Intravenous cannulation in children
- Laser skin treatments
- Lithotripsy
- Circumcision

EMLA cream should not be used on mucous membranes, broken skin, infants younger than 1 month of age, or patients with a predisposition to methemoglobinemia. EMLA is applied 45-60 minutes before the procedure with an occlusive dressing. It provides analgesia up to a depth of 5 mm, which lasts for 1-2 hours.

Other side effects include- blanching of the skin, erythema, and oedema.

Solution to Question 3:

The image shows a nasopharyngeal airway and its insertion. The true statement is that it can be used in conscious patients.

A nasopharyngeal airway is used to provide artificial airway passage to the hypopharynx. To place a nasopharyngeal airway, the device should be lubricated and inserted into the nose with the bevel facing the nasal septum, at a 90-degree angle to the longitudinal axis of the body. It bypasses the gag reflex as they do not go through the mouth. Thus, it can be used in conscious patients or those who have been lightly anesthetized.

Care should be taken to avoid excessive force as damage to the nasal mucosa can result in epistaxis. The nasal approach should be avoided in conditions that put patients at higher risk for epistaxis. These conditions include but are not limited to coagulopathy, pregnancy, hereditary hemorrhagic telangiectasia, and uncontrolled hypertension. Absolute contraindications include basilar skull fractures, facial trauma, midface disruption, and injury to the roof of the mouth.

An oropharyngeal airway is used to maintain the airway in anesthetized patients. This adjunct prevents the tongue and epiglottis from falling back against the posterior pharynx and blocking

the airway. They can cause coughing, retching, or laryngospasm in conscious patients.

To place an oropharyngeal airway, the device is inserted with the curved portion facing downwards. It is then rotated 180 degrees to be put in place. The below image shows the oropharyngeal airway:



Solution to Question 4:

Tracheostomy would be difficult to perform in this case of pneumothorax with tracheal deviation.

The trachea is displaced from its usual midline position, making it challenging to accurately locate and access the trachea for the incision. The tracheostomy procedure requires precise placement to ensure proper airway management and avoid complications.

Steps of Tracheostomy:

- **Incision:** Make a vertical incision in the midline from the cricoid cartilage to above the sternal notch. For elective procedures, a 5 cm transverse incision two fingers' breadth above the sternal notch can be used for a better cosmetic outcome.
- **Dissection:** Dissect tissues in the midline. Displace or ligate any dilated veins encountered to control bleeding.
- **Muscle Separation:** Separate and retract the strap muscles of the neck laterally to expose underlying structures.
- **Thyroid Isthmus Management:** Displace the thyroid isthmus upwards or divide and ligate it with sutures.
- **Local Anesthesia:** Inject a few drops of 4% lignocaine into the trachea to suppress coughing.
- **Tracheal Incision:** Fix the trachea with a hook and make a vertical incision between the third and fourth tracheal rings, converting it into a circular opening. Avoid dividing the first tracheal ring to prevent perichondritis and stenosis.

- **Inserting the Tube:** Insert a tracheostomy tube of appropriate size into the tracheal opening and secure it with tapes.
- **Avoiding Tight Suturing:** Avoid tightly suturing or packing the skin incision to prevent subcutaneous emphysema.
- **Dressing:** Place a gauze dressing between the skin and the flange of the tracheostomy tube around the stoma to absorb secretions, provide a barrier against infection, and allow air to escape, promoting proper healing

Other Options:

Option A: Nasogastric tube insertion: Targets the esophagus, so tracheal deviation has minimal impact.

Option B: Laryngeal mask airway (LMA): Positioned over the larynx, not directly affected by tracheal deviation.

Option D: Indirect laryngoscopy: Focuses on the larynx and vocal cords, and is less influenced by the trachea's position.

Solution to Question 5:

Sugammadex is used for immediate reversal of vecuronium-induced neuromuscular blockade/relaxant in the given scenario.

Sugammadex is a modified γ -cyclodextrin and the first selective relaxant-binding agent based on an encapsulating principle for the inactivation of nondepolarising neuromuscular blocking drugs mainly rocuronium and vecuronium. This principle of complex formation of sugammadex and rocuronium or vecuronium in a 1:1 ratio occurs at all levels of neuromuscular blockade resulting in a more fast-acting pharmacologic reversal when compared with anticholinesterase drugs like neostigmine.

Other options:

Options A and B: Physostigmine and Rivastigmine are acetylcholinesterase inhibitors, which are not effective for reversing vecuronium.

Option D: Succinylcholine is a neuromuscular blocker itself and would not reverse vecuronium; it is used to induce paralysis rather than reverse it.

Solution to Question 6:

This patient in the above clinical scenario has developed a sudden onset of hypoxia, decreased breath sounds on one side, and increased intrathoracic pressure, along with an X-ray showing hyper lucency in the left side with tracheal and mediastinal shift to the right side is suggestive of a diagnosis of Pneumothorax.

A pneumothorax occurs when air enters the pleural space, causing lung collapse and preventing normal lung expansion, which leads to absent or diminished breath sounds over the affected area. Pneumothorax can cause a rapid increase in intrathoracic pressure, which impairs lung expansion

and leads to significant hypoxia. In patients on mechanical ventilation, it delivers positive pressure to the lungs, which can exacerbate or reveal underlying pleural issues. Increased positive pressure can contribute to the development or worsening of a tension pneumothorax.

Other options:

Option B: Left-Sided Pleural Effusion: While this condition can also cause decreased breath sounds and hypoxia, it typically does not cause acute increases in intrathoracic pressure.

Option C: Acute Respiratory Distress Syndrome (ARDS): ARDS is characterized by diffuse lung inflammation and can lead to hypoxia. However, it generally does not cause localized decreased breath sounds.

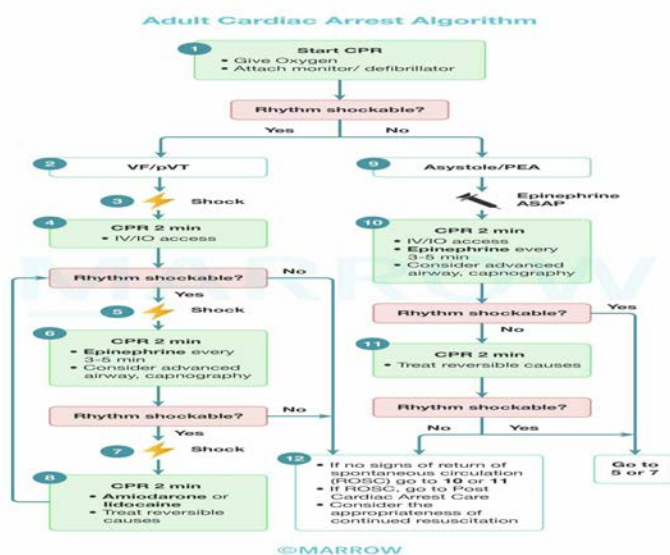
Option D: Left-Sided Pneumonia: Pneumonia can cause hypoxia and decreased breath sounds but is less likely to be associated with a sudden increase in intrathoracic pressure.

Solution to Question 7:

The given ECG showing polymorphic ventricular tachycardia, which is a shockable rhythm, in a stuporous paediatric patient warrants immediate defibrillation followed by CPR for 2 minutes.

Polymorphic ventricular tachycardia, especially in a paediatric patient, is often associated with severe underlying conditions and is a life-threatening rhythm disturbance. Immediate defibrillation is critical for restoring a normal rhythm.

After defibrillation, it is advisable to do CPR for 2 minutes before reassessing the rhythm for the next round of shock to ensure adequate blood circulation and oxygenation. .

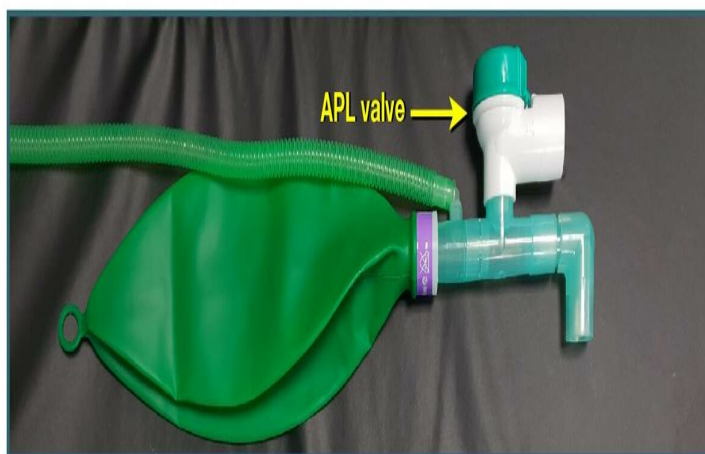


Note: Cardioversion is not suitable for polymorphic VT; defibrillation is the preferred approach in such cases.

Solution to Question 8:

In the image, the marked structure is the adjustable pressure-limiting (APL) valve of Mapelson-C anaesthetic breathing system and it will open at the pressure of 30-40 cm H₂O. This valve is designed to release gas when a certain pressure is exceeded to prevent excessive pressure from building up in the patient's airway.

The APL valve typically opens at pressures of 30-40 cm H₂O to allow for safe ventilation without causing barotrauma to the lungs. This range is appropriate for manual ventilation to prevent high airway pressures that could lead to lung injury.



Solution to Question 9:

The position illustrated in the picture is known as the recovery position, and it is used in the management of patients who are unresponsive but with pulse present and breathing present.

The recovery position can help prevent complications such as aspiration and airway obstruction in unconscious individuals. Additionally, it is crucial to continuously monitor the patient's airway, breathing, and circulation while in recovery and seek immediate medical assistance. An unresponsive patient who is breathing spontaneously in the supine position is at risk of airway obstruction from their tongue, mucus, vomit, etc. A modified lateral position is used and the following 6 principles are considered while choosing which variation of recovery position to use,

-

The victim should be in as near a true lateral position as possible, with the head dependent to allow free drainage of fluid.

-

The position should be stable.

-

Avoid any pressure on the chest that impairs breathing.

-

It should be possible to turn the victim on his or her side and to return to the back easily and safely, with concern for a possible cervical spine injury.

-

Good observation of and access to the airway should be possible.

-

The position itself should not cause injury to the victim.

Other options:

Option A: Unresponsive with pulse and breathing absent - In this situation, the patient is in cardiac arrest and requires immediate cardiopulmonary resuscitation (CPR), not the recovery position. CPR involves chest compressions and rescue breaths to support circulation and respiration.

Option B: Responsive patient with pulse and breathing present- The patient should be monitored until the emergency response team arrives.

Option D: Unresponsive with pulse present and breathing absent - In this case, the patient is experiencing respiratory arrest. Rescue breathing or bag-valve-mask ventilation should be initiated to support the patient's breathing. The recovery position is not indicated in this scenario.

Solution to Question 10:

In a capnographic waveform, the plateau phase corresponds to alveolar air and plateau pressure.

The normal waveform has four different phases:

Phase I: Inspiratory baseline, represents the CO₂-free gas from the airways (anatomical and apparatus dead space).

Phase II: Expiratory upstroke, mixing of dead space gas with alveolar gas.

Phase III: Alveolar plateau, represents CO₂-rich gas from the alveoli. The last of the alveolar gas is sampled to measure ET-CO₂.

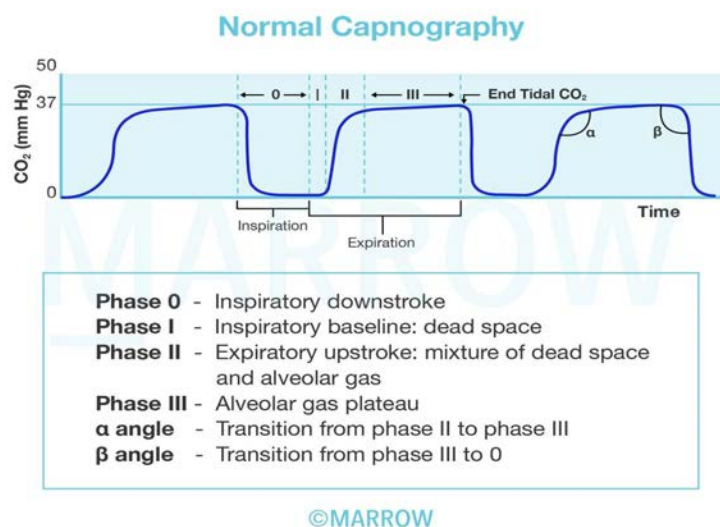
Phase o: Inspiratory downstroke and the beginning of inspiration.

Alpha angle: The transition from phase II to III.

The alpha angle can be used to assess the ventilation/perfusion of the lung. V/Q mismatches will have an alpha angle greater than 90 degrees.

Beta angle: The transition from phase III to o.

The beta angle can be used to assess rebreathing. If rebreathing occurs, the angle is greater than 90 degrees.



Solution to Question 11:

The position in the image of a hand with a tourniquet occluded limb shows a Bier Block.

IVRA or Bier's block involves the intravenous administration of a local anesthetic into a tourniquet-occluded limb. It is used to provide anesthesia for short surgical procedures (<90 minutes) on the limbs.

Other options:

Option B: The axillary nerve block provides anesthesia for short surgical procedures on the hand, wrist, or forearm. It is performed by using a ultrasound-guided needle in the proximal axilla between the tendon of the latissimus dorsi and the teres major.

Option C: Median nerve blocks are used for procedures involving the palmar aspect of the arm or the carpal tunnel. For a median nerve block at the elbow, the probe is placed in the elbow crease over the proximal radius and the anesthetic is injected aiming for the fascial plane between flexor digitorum superficialis and flexor digitorum profundus.

Option D: For a wrist block, the arm is abducted with the hand in supination, following which the anesthetic is infiltrated in the dorsal aspect of the wrist. This is used for short procedures of the wrist and the fingers.

Note: Bupivacaine is not used for Bier's block. It is the most cardiotoxic local anesthetic.

Accidental early deflation of the tourniquet can lead to massive systemic doses of bupivacaine resulting in cardiac arrhythmias and cardiovascular collapse.

Solution to Question 12:

The above image shows the sniffing position used for intubation, which is the true statement.

The sniffing position is a common head and neck alignment technique used during intubation to optimize the view of the glottis for easier insertion of the endotracheal tube. It is achieved by flexing the neck and extending the head, resembling the posture one assumes when sniffing. The position aligns the oral, pharyngeal, and laryngeal axes to provide a straight line of sight, making the glottis more visible during direct laryngoscopy and intubation.

Steps of intubation using the sniffing position:

- Direct laryngoscopy is performed in the sniffing position, with the head elevated 5-10 cm above the surgical table, the cervical spine flexed, and the atlantooccipital joint extended.
- The epiglottis is lifted up by the tip during laryngoscopy with Miller's blade.
- While performing laryngoscopy, the scope is lifted upwards and never levered over the upper incisors.

Other options:

Option A: The sniffing position is the most commonly used position in anesthesia as it ensures patient safety by ensuring a straight line of sight, allowing for better outcomes of the procedure.

Option B: The sniffing position is not the ideal position for bougie insertion during emergencies as there is concerns for cervical spine trauma in patients in the emergency department.

Option C: The sniffing position is not the typical position for neurosurgery. Depending on the location of the surgery, different head and neck positions are used.

Solution to Question 13:

This patient presenting with stridor after prolonged intubation most likely has subglottic stenosis post intubation. This condition is staged by the Cotton Meyer's grading.

Subglottic stenosis following prolonged intubation is a narrowing of the airway just below the vocal cords (subglottic region), which can lead to difficulty breathing. This condition is often a complication of endotracheal intubation due to the pressure exerted by the tube or cuff on the airway mucosa, leading to injury and scarring.

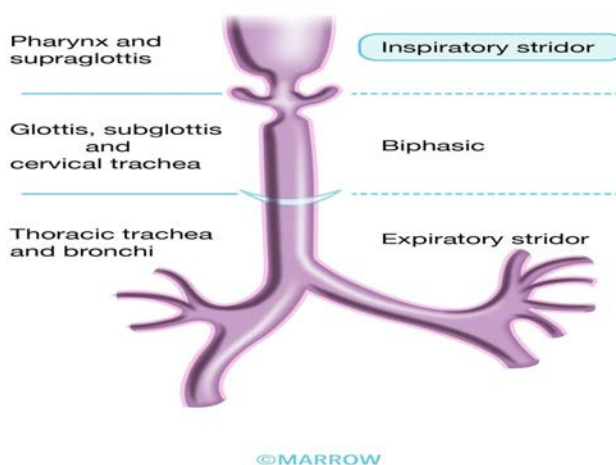
These patients present with stridor, dyspnea, hoarseness or changes in voice, and difficulty in breathing.

Diagnosis is by laryngoscopy or bronchoscopy which allows direct visualization of the subglottic area to assess the degree of stenosis.

CT or MRI may be used to evaluate the extent of narrowing and identify any underlying structural damage.

Treatment can be via endoscopic balloon or mechanical dilation, laser therapy or surgical resection may be needed to remove the stenosis in severe cases. In severe cases, a tracheostomy may be required to bypass the obstructed segment and allow for breathing.

Types of stridor and their site of origin



Solution to Question 14:

In the clinical scenario given above, the development of tachycardia and muscle rigidity soon after administration of anesthetic agents can be due to malignant hyperthermia. Halothane is the anesthetic agent associated with this condition.

Malignant hyperthermia is a rare genetic hypermetabolic disease of the muscle. It occurs due to a mutation of the gene for the ryanodine receptor (Ryr 1), located on chromosome 19. It is an ion channel responsible for calcium release from the sarcoplasmic reticulum.

The symptoms develop when the patient is exposed to suxamethonium/succinylcholine or other inhaled anesthetics like halothane, isoflurane, sevoflurane, desflurane, etc. The mutated ryanodine receptor leads to a sudden release of calcium from the sarcoplasmic reticulum of the skeletal muscle. This uncontrolled increase in intracellular calcium removes the inhibition of troponin, resulting in sustained muscle contraction, and increasing adenosine triphosphatase (ATPase) activity. This results in an uncontrolled hypermetabolic state with increased oxygen consumption and CO₂ production, producing severe lactic acidosis and hyperthermia.

The key clinical features are:

- Succinylcholine-induced masseter muscle rigidity (causing trismus) or generalized muscle rigidity
- Tachycardia
- Hypercarbia (increased CO₂ production) and reduced oxygen saturation
- Hyperthermia (rise of 1° Celsius can occur every 5 minutes)
- Arrhythmia
- Hypertension
- Myoglobinuria and myoglobinemia indicating muscle damage
- Metabolic derangements like hyperkalemia and mixed metabolic and respiratory acidosis

Treatment is to discontinue the triggering agents i.e. succinylcholine followed by administration of 100% oxygen. Metabolic acidosis and hyperkalemia need to be corrected. Hyperthermia must be controlled by cooling the patient. Dantrolene sodium is the drug of choice. It is given at a dose of 2.5 mg/ kg intravenously and can be repeated every 5-10 minutes till the attack subsides.